

From Data to Digital Twins: The Future of Asset Management

Presented by: Grace Sammartino
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August 2026



From Data to Digital Twins: The Future of Asset Management

Agenda

- Company Overview
- Geospatial Digital Delivery
- Digital Twin
- Showcase



Presented by: Grant
Account Executive

A world map with a dark grey background. Numerous small dots, primarily in shades of orange and yellow, are scattered across the landmasses, representing the global distribution of Esri users. The dots are most densely packed in North America, Europe, and East Asia.

esri

World Leader in Geographic Information Systems (GIS)

Over 10 Million Users

33,000 Businesses

12,000 Colleges & Universities

8,600 Utilities

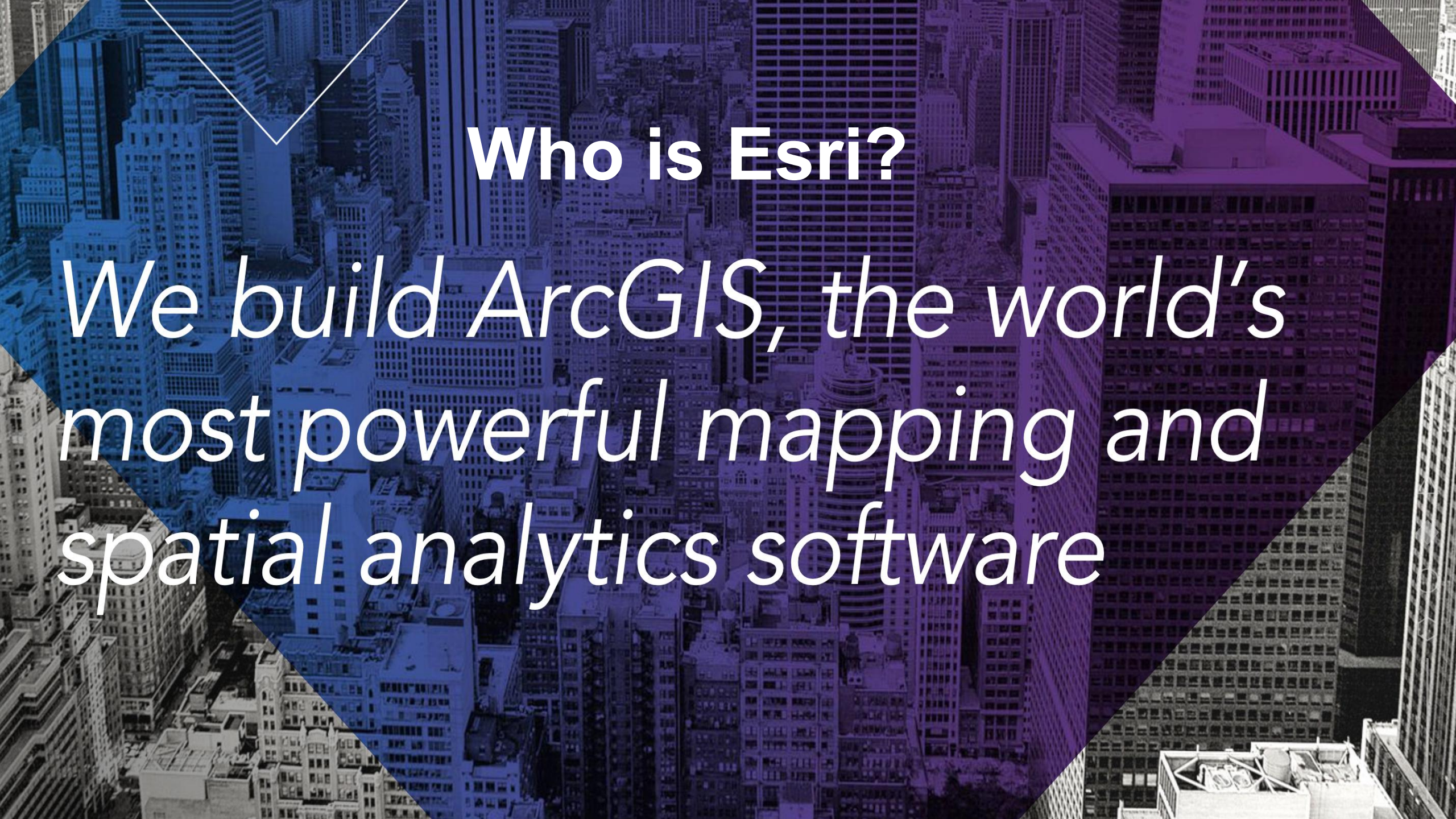
12,600 NGOs

1,200 National Government Agencies

12,000 States & Regional Agencies

30,800 Cities & Local Governments

Empowering Over 680K Organizations Worldwide



Who is Esri?

*We build ArcGIS, the world's
most powerful mapping and
spatial analytics software*

ArcGIS

Data Fusion Platform

Dashboards

Performance
Management

IoT

Autonomous
Vehicles

Real-Time

Analytics

Situational
Awareness

Sensors

Artificial
Intelligence

Environmental
Monitoring

Machine-Learning



What Is a GIS?

A **System** for Managing, Sharing and **Synthesizing** Geographic Information



What Is a GIS?

The Original Digital Twin

A **System** for Managing, Sharing and Applying Geographic Information



ArcGIS is a System of Systems

Connecting Department of Transportations for Common Goals



Action

Understanding

Knowledge

Asset Management

Capital Projects

Traffic Operations

Safety

Maintenance Operations

Incident Management

Planning & Performance

Environmental

Multimodal

Digital Delivery

Equity & Access

Climate Resilience

Predictive Analytics

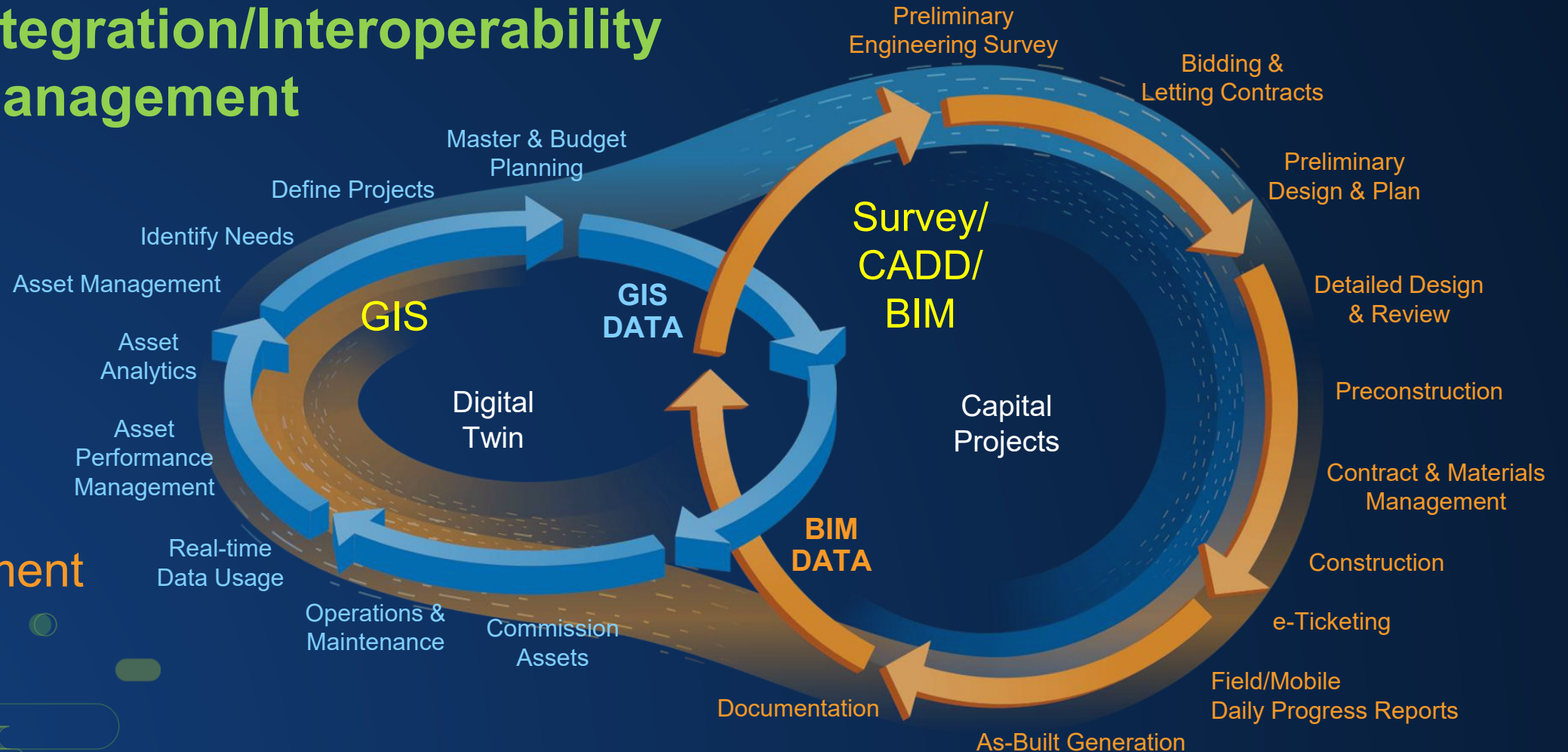
...Enabled by AI-ready enterprise GIS and integrated data.

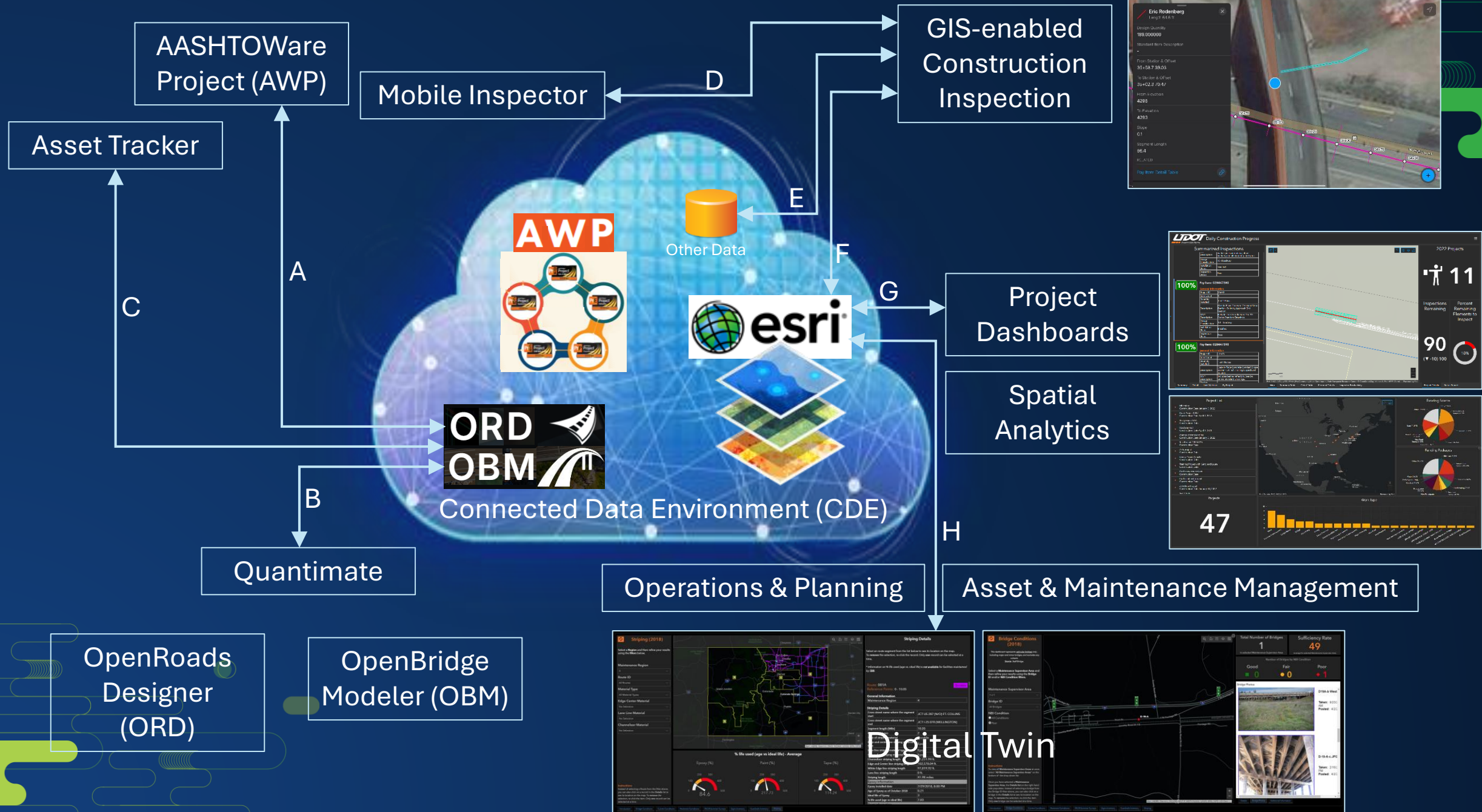
Digital Delivery

Moving Asset Location and Descriptive Data Seamlessly with Zero Loss Throughout Its Lifecycle

Data Governance/Data Modeling
System Integration/Interoperability
Change Management

**Asset
Management
Lifecycle**





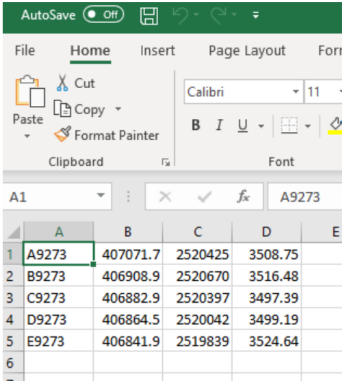
Data Model Changes in Design & Survey

Spatial Representation (graphics) and Attributes were often separated between toolsets & manually linked

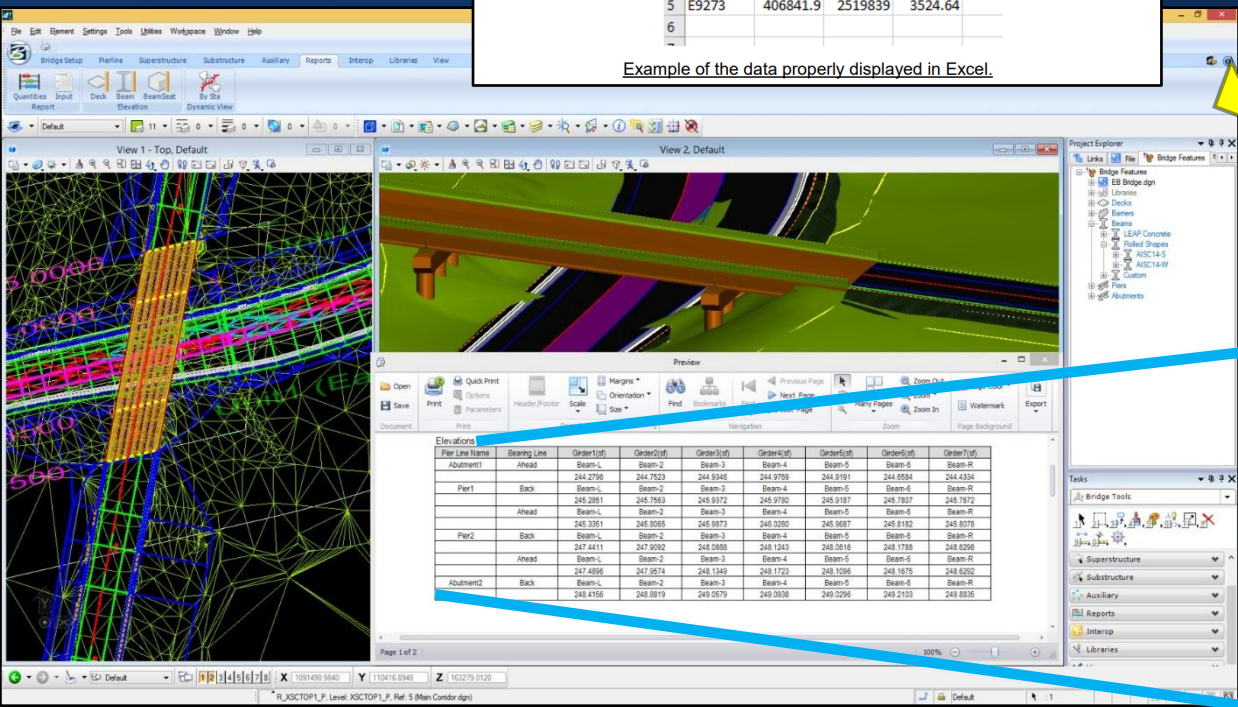
Feature Codes linked to surveyed spatial representations but not to design elements

Design Environment storing spatial representation

7. Save the file by clicking the **Save As** button and navigating to the proper location. In the file type drop-down window, select **CSV (comma delimited).csv** and click **Save**.



Example of the data properly displayed in Excel.



Pay/Bid Items linked manually to spatial representation or used just in tabular form

Final Estimation Bridge 5-3-11 [Compatibility Mode] - Excel

ESTIMATION FOR FABRICATION AND INSTALLATION OF PEDESTRIAN STEEL TRUSS BRIDGE ON 4 LANE
BORDER WITH SERVICE ROAD AND DRAINS IN THE BUILT UP AREA (KM 11.88 TO KM 28.33) LENGTH 16.36

FDE-C&W-111-000 Length-37380 mm

BILL NO. 4.4 STRUCTURES (STEEL PEDESTRIAN)

BILL OF QUANTITY

Part No.	Item Tag	Description	Material Specs	Unit	TYP	L (mm)	W (mm)	T (mm)	L (mm)
01	Foundations & Columns	Anchor Bolts	JIS G3101 S5 400	Kgs.	High Strength Bolts	Ø 25x1000			
02	Foundations & Columns	Hex. Nuts & Washers	A 325	Kgs.	High Strength Nuts	M24*A26			
03	Foundations & Columns	Templates	HR 235	Kgs.	Plate	500	x	500	x
04	Foundations & Columns	Base Plates	HR 235	Kgs.	Plate	500	x	500	x
05	Foundations & Columns	Rib Plates	HR 235	Kgs.	Plate	500	x	300	x
06	Foundations & Columns	Rib Plates	HR 235	Kgs.	Plate	150	x	300	x
07	Foundations & Columns	Main Column	ASTM A-36 (Hot Rolled)	Kgs.	H- Beam	250	x	250	x
08	Foundations & Columns	Stair Coolumn	ASTM A-36 (Hot Rolled)	Kgs.	H- Beam	250	x	250	x
09	Foundations & Columns	Top Joint Plate	HR 235	Kgs.	Plate	600	x	300	x
10	Foundations & Columns	Top Rib	HR 235	Kgs.	Plate	250	x	200	x
11	Foundations & Columns	Top Hex. Head Bolts+Nuts	A 325	Nos.	Nut+Bolts	M24 x 100			
01	Column Bracing	Angle	ASTM A-36 (Hot Rolled)	Kgs.	Angle	100	x	100	x
02	Column Bracing	Angle	ASTM A-36 (Hot Rolled)	Kgs.	Angle	100	x	100	x
03	Column Bracing	Angle	ASTM A-36 (Hot Rolled)	Kgs.	Angle	100	x	100	x
04	Column Bracing	Angle	ASTM A-36 (Hot Rolled)	Kgs.	Angle	100	x	100	x
05	Column Bracing	Gusset Plate	HR 235	Kgs.	Plate	400	x	200	x
06	Column Bracing	Gusset Plate	HR 235	Kgs.	Plate	350	x	300	x
07	Column Bracing	Gusset Plate	HR 235	Kgs.	Plate	200	x	200	x
01	Truss	Main Boom Angle	ASTM A-36 (Hot Rolled)	Kgs.	Angle	175	x	175	x
02	Truss	Top Angle	ASTM A-36 (Hot Rolled)	Kgs.	Angle	175	x	175	x
03	Truss	Vertical Angle	ASTM A-36 (Hot Rolled)	Kgs.	Angle	100	x	100	x

Specific Engineering Data from the Design Environment

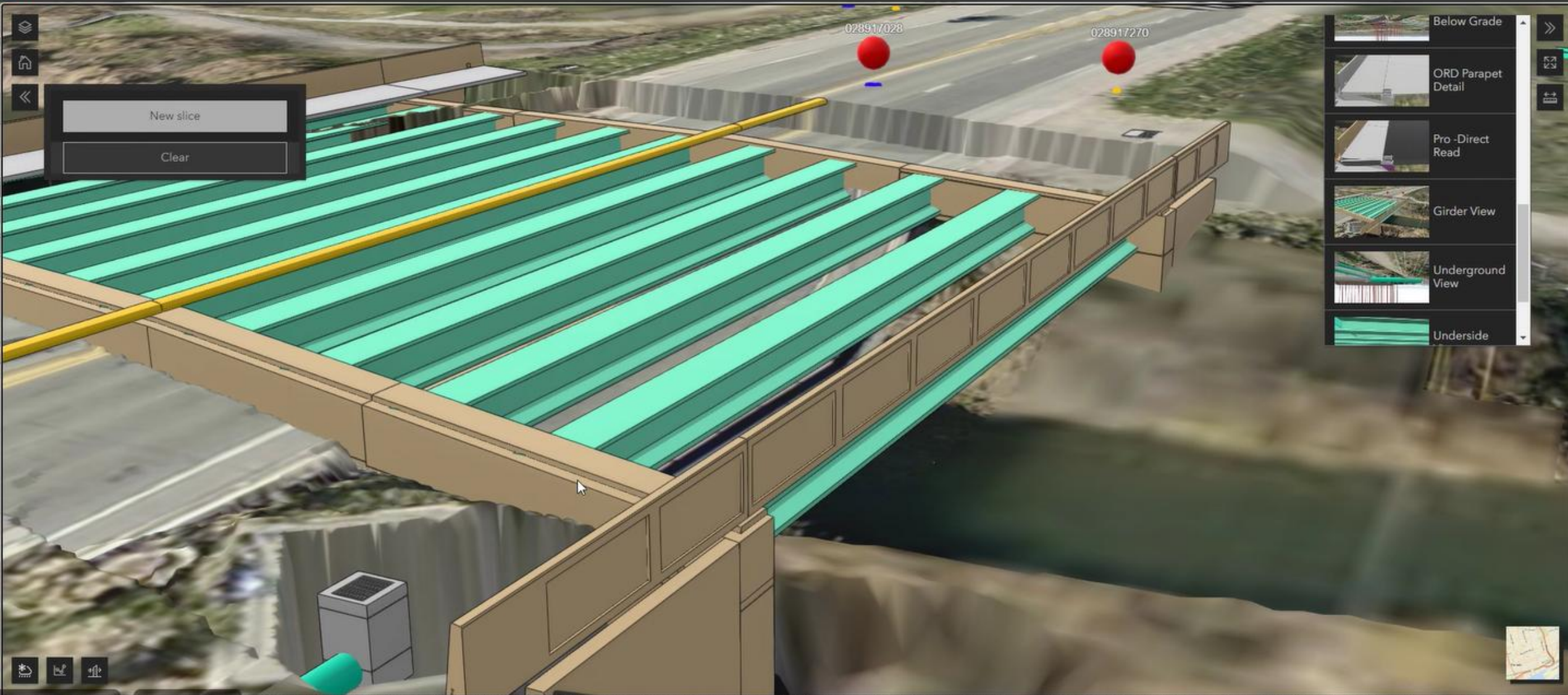
Elevations

Pier Line Name	Bearing Line	Girder1(sf)	Girder2(sf)	Girder3(sf)	Girder4(sf)	Girder5(sf)	Girder6(sf)	Girder7(sf)
Abutment1	Ahead	Beam-L	Beam-2	Beam-3	Beam-4	Beam-5	Beam-6	Beam-R
		244.2796	244.7523	244.9346	244.9769	244.9191	244.6584	244.4334
Pier1	Back	Beam-L	Beam-2	Beam-3	Beam-4	Beam-5	Beam-6	Beam-R
		245.2851	245.7563	245.9372	245.9780	245.9187	245.7807	245.7672
	Ahead	Beam-L	Beam-2	Beam-3	Beam-4	Beam-5	Beam-6	Beam-R
		245.3351	245.8065	245.9873	246.0280	245.9687	245.8182	245.8078
Pier2	Back	Beam-L	Beam-2	Beam-3	Beam-4	Beam-5	Beam-6	Beam-R
		247.4411	247.9092	248.0868	248.1243	248.0616	248.1788	248.6298
	Ahead	Beam-L	Beam-2	Beam-3	Beam-4	Beam-5	Beam-6	Beam-R
		247.4896	247.9574	248.1349	248.1723	248.1096	248.1675	248.6292
Abutment2	Back	Beam-L	Beam-2	Beam-3	Beam-4	Beam-5	Beam-6	Beam-R
		248.4156	248.8819	249.0579	249.0938	249.0296	249.2103	249.8835

Moving as-built data to asset management & digital twin

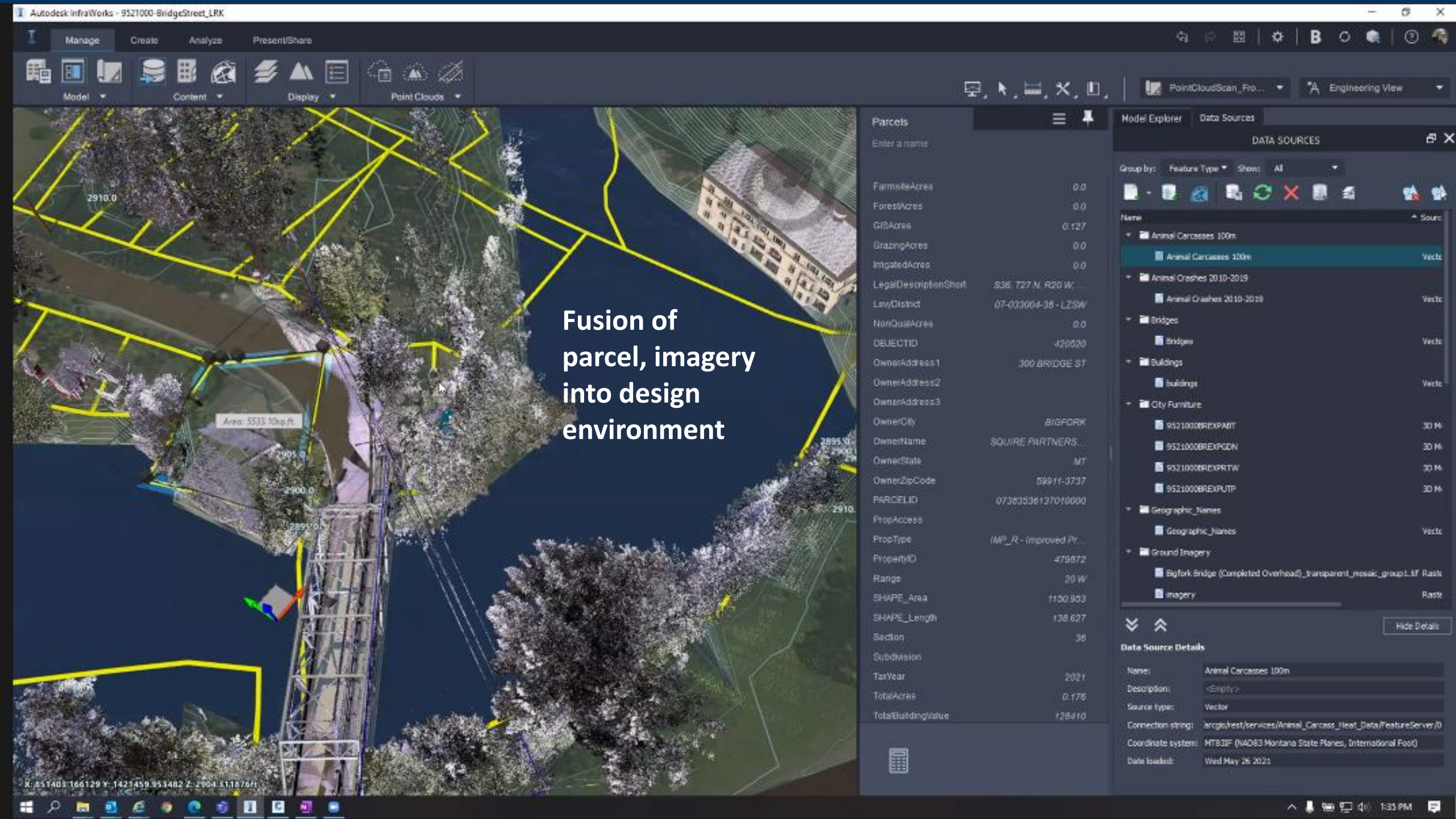


Daily Construction Progress



Existing GIS Data can be integrated, visualized and analyzed in the Digital Twin including Linear Referencing System (LRS) technology





Fusion of
parcel, imagery
into design
environment

Parcels

Enter a name

FarmSiteAcres	0.0
ForestAcres	0.0
GIRAcres	0.127
GrazingAcres	0.0
IrrigatedAcres	0.0
LegalDescriptionShort	S36, T27 N, R20 W, ...
LevyDistrict	07-033004-38 - LZSW
NonQualAcres	0.0
OBJECTID	426020
OwnerAddress1	300 BRIDGE ST
OwnerAddress2	
OwnerAddress3	
OwnerCity	BIGFORK
OwnerName	SQUIRE PARTNERS...
OwnerState	MT
OwnerZipCode	09911-3737
PARCELID	07363336137010000
PropAccess	
PropType	IMP_R - Improved Pr...
PropertyID	479872
Range	20 W
SHAPE_Area	1150.933
SHAPE_Length	138.627
Section	36
Subdivision	
TaxYear	2021
TotalAcres	0.176
TotalBuildingValue	128410

Model Explorer

Data Sources

DATA SOURCES

Group by:	Feature Type	Show:	All
Name			Source
Animal Carcasses 100m			Vector
Animal Crashes 2010-2019			Vector
Bridges			Vector
Buildings			Vector
City Furniture			3D Model
Geographic Names			Vector
Ground Imagery			Raster



Data Source Details

Name:	Animal Carcasses 100m
Description:	<Empty>
Source type:	Vector
Connection string:	arcgis/rest/services/Animal_Carcass_Heat_Data/FeatureServer/0
Coordinate system:	MTB31F (NAD83 Montana State Planes, International Foot)
Date loaded:	Wed May 26 2021

An aerial view of a city with a digital twin overlay. The overlay shows a large, white, curved structure with a grid of points, likely representing a digital twin of a building or infrastructure. The text "From Concept to Reality" is in white, and "Building Living Digital Twins with Enterprise GIS" is in yellow and red. The background is a dark, blue-tinted aerial view of a city with various buildings and infrastructure.

From Concept to Reality

Building Living Digital Twins with Enterprise GIS

What is a Digital Twin?

A data-driven virtual representation of real-world entities and processes, synchronized at a specified frequency and fidelity.

Understanding, Decision-Making & Action

Transform businesses by accelerating and automating holistic understanding, optimal decision-making, and effective action.

Outcomes, Use cases, Integrations

Are motivated by outcomes, tailored to use cases, powered by integration, built on data, guided by domain knowledge, and implemented in IT/OT/ET systems.

Past, Current & Future

Use real-time and historical data to represent the past, present and simulate predicted futures.

Digital Twin Purpose

Connecting People and Systems
for a **Shared Understanding**...

*...to make better decisions
and drive real-world outcomes*



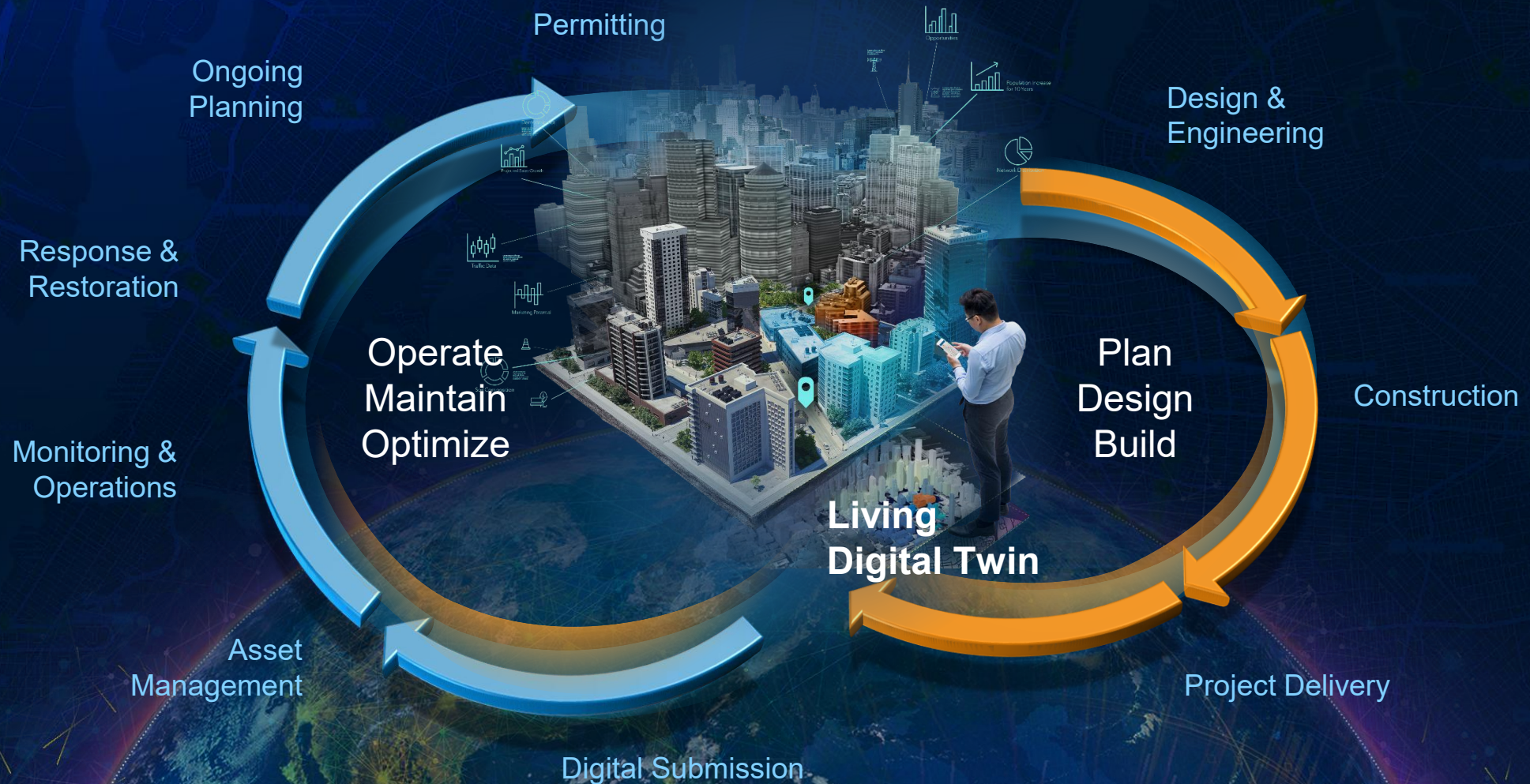
Digital Twin

Location is Foundational for Digital Twins



GIS

Supports Complete Infrastructure Digital Twin Lifecycle



Digital Twin

Maturity



01. DESCRIPTIVE

Virtual representations of the physical assets or networks

Reality Mapping

Virtual Model (Static)

2D/3D Data

02. INFORMATIVE

Real-time connection to sensors from multiple sources

IoT Sensors

Analytics

Connected Data Environments

Virtual Model (Dynamic)

03. PREDICTIVE

Capable of predicting outcomes

AI / ML

Forecasting & Simulations

Advanced Analytics

04. COMPREHENSIVE

Scenarios of outcomes to make recommendations

Scenario Based Modeling

Remote Controlled Inspections

05. AUTONOMOUS

Self-learning and decision-making with limited human interaction

Autonomous Operations

Adaptive

AI Assist & Decision

Living Digital Twin



ArcGIS powers outcome-driven digital twin

Showcase



ENVIRONMENTAL TWIN

- Monitor environmental conditions
- Scenario modeling
- Environmental vulnerabilities (e.g., heat, flooding, drought, air quality)
- Proactive mitigation planning.
- Integrates sensor, imagery, and simulation data
- Communicates complex environmental insights clearly to stakeholders.



DESIGN & CONSTRUCTION TWIN

- 3D Design and Planning
- Integrates BIM, GIS, schedules, and field data into a unified, accurate project view.
- Improves coordination and reduces risk through spatial clash detection and sequencing.
- Real time updates for field inspections
- Optimizes cost, schedule, and sustainability with data-driven insights.
- Captures as-built data to improve Digital Delivery



NETWORK TWIN

- Connectivity Model & Network Topology
- Complex Network Representations
- Optimizes Operational efficiency
- Real Time Network Analysis
- Improves Data Quality and Governance
- Single Source of truth for the Network
- Pervasive access from field to office.



PLANNING TWIN

- Evaluate land-use, zoning, and development impacts
- Data-driven scenario planning
- Establish KPI dashboards for evaluation
- Improves equity analysis by evaluating community-level impacts.
- Enhances public engagement
- Visual storytelling of future plans and developments.



EMERGENCY MANAGEMENT TWIN

- Plan, Respond and Recover
- Common Operating Picture
- Rapid Damage Assessment
- Geo AI & ML
- Customer & Community Engagement



OPERATIONS TWIN

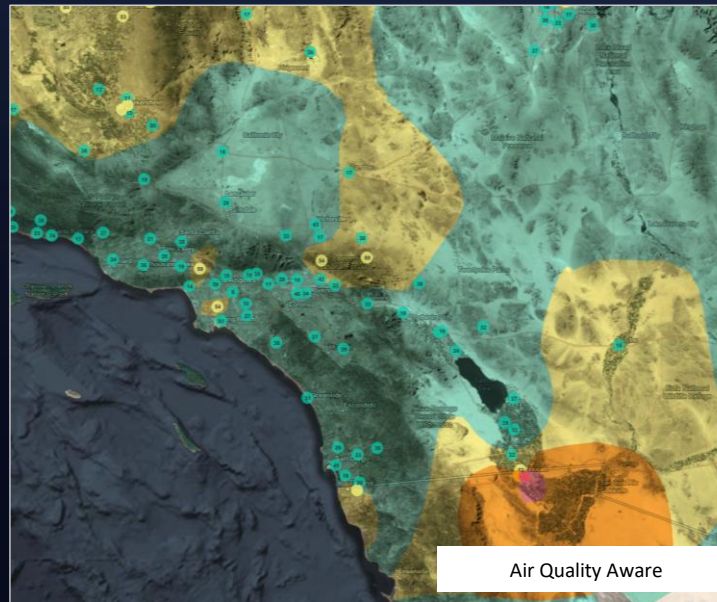
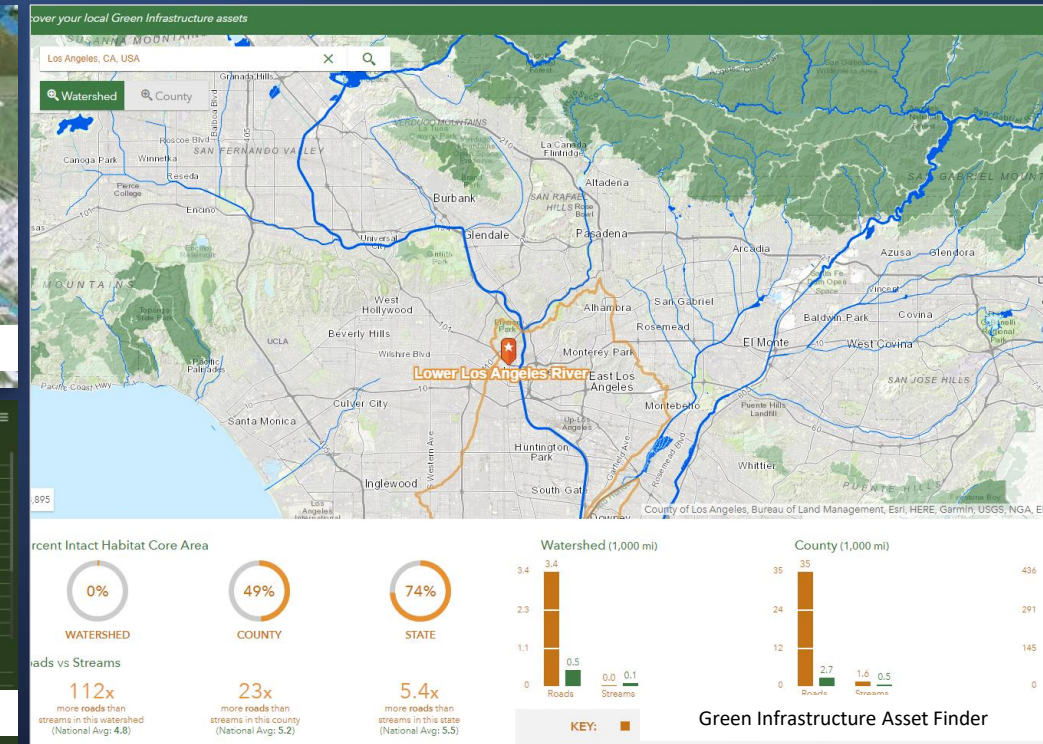
- Real-Time visibility
- Situation Awareness
- Rapid response and incident management
- Optimize space utilization, traffic flow, energy usage, and service delivery.
- Connect operational systems to a common geospatial frame.
- Predictive insights, simulations, and automated workflows.



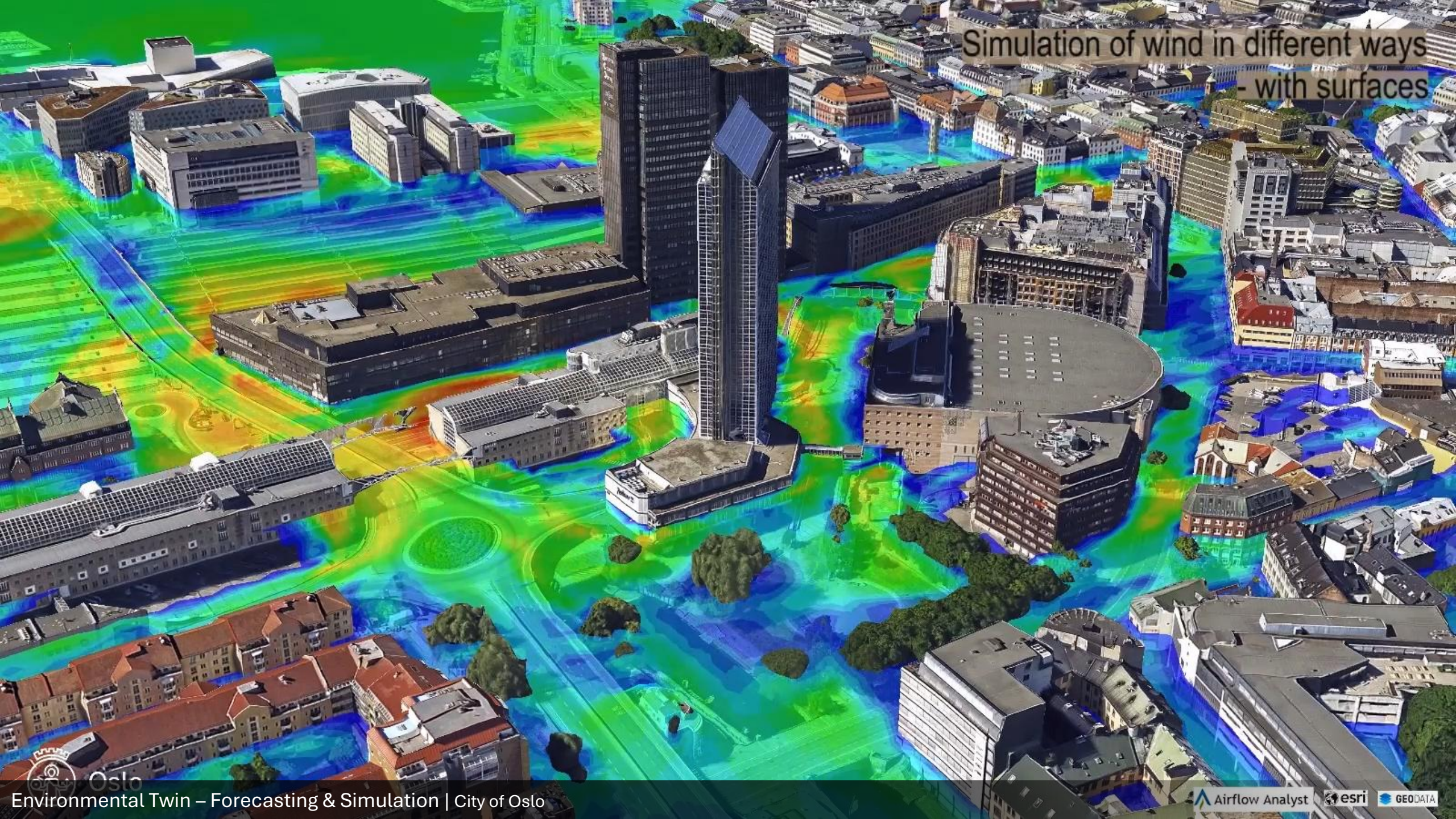


Environmental Twin

- Provides a unified spatial environment to monitor climate, ecological, and natural resource conditions in real-time.
- Enables scenario modeling to understand impacts of environmental vulnerabilities (e.g., heat, flooding, drought, air quality)
- Supports proactive mitigation planning.
- Integrates sensor, imagery, and simulation data to improve forecasting and early-warning capabilities.
- Communicates complex environmental insights clearly to stakeholders.



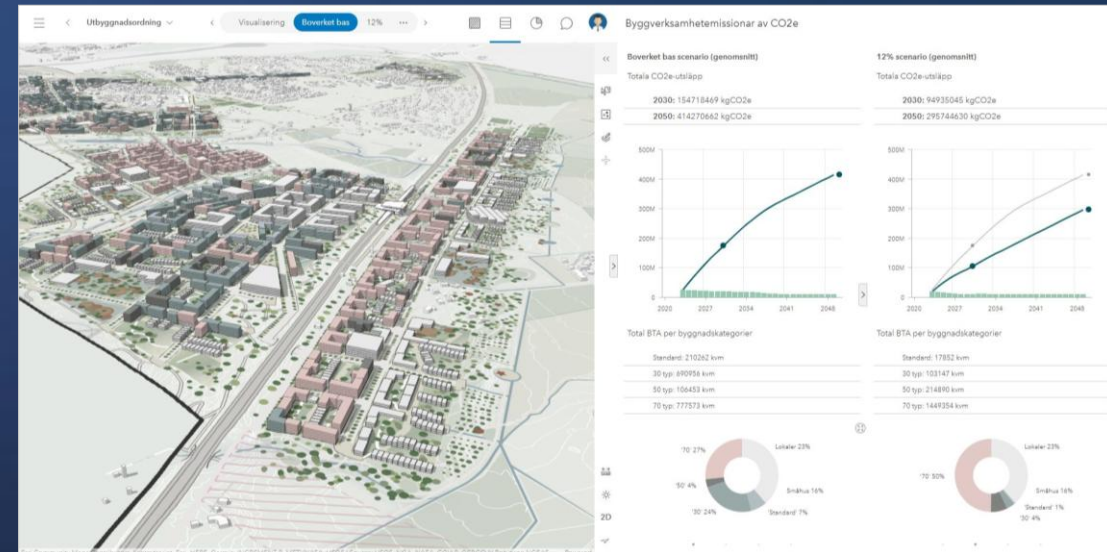
Simulation of wind in different ways
- with surfaces

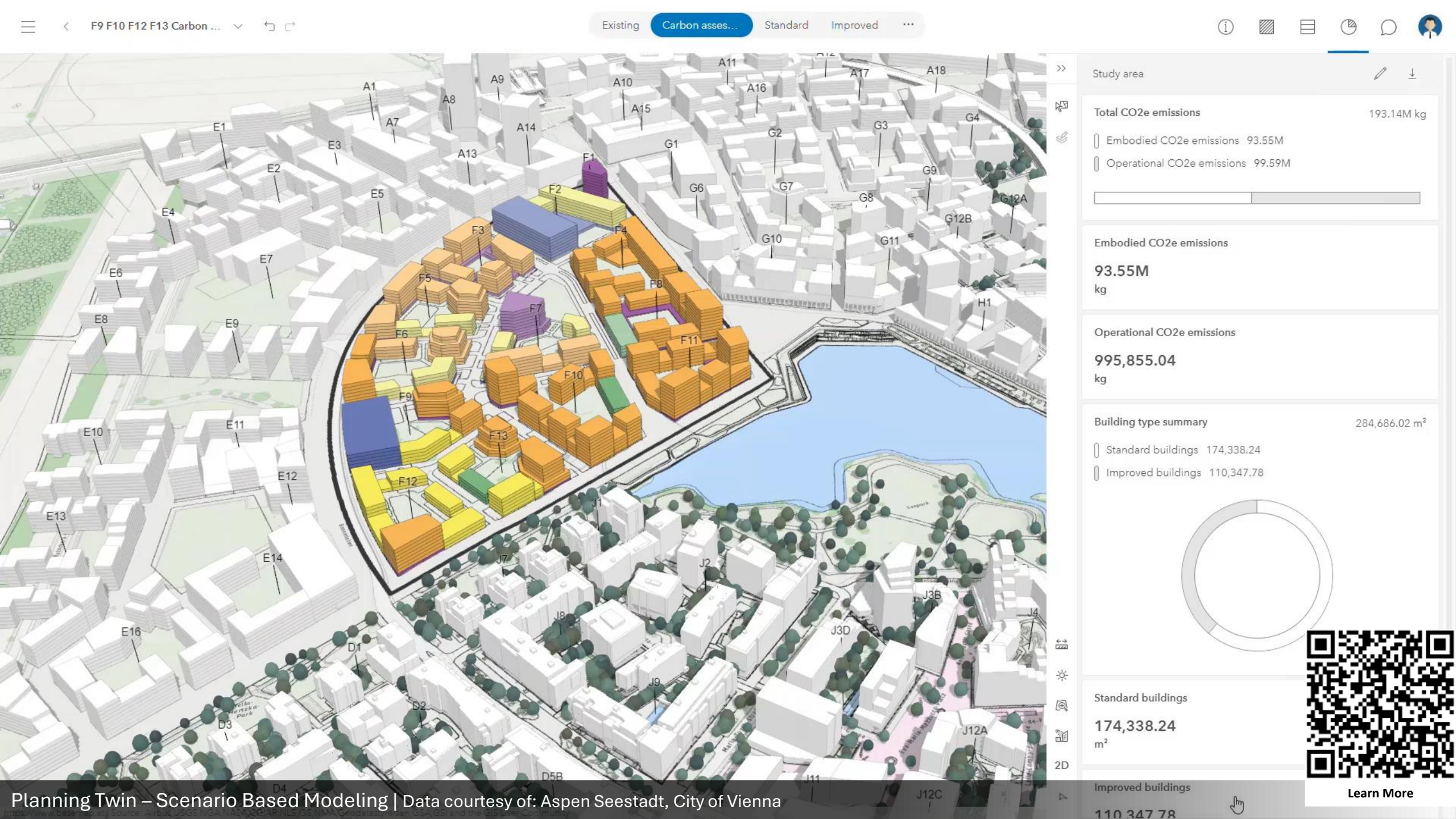




Planning Twin

- Evaluate land-use, zoning, and development impacts.
- Enable data-driven scenario planning based on defined strategies.
- Establish KPI dashboards for evaluation (accessibility, financial performance, housing, etc.)
- Improves equity analysis by evaluating community-level impacts.
- Enhances public engagement through interactive, visual storytelling of future plans and developments.

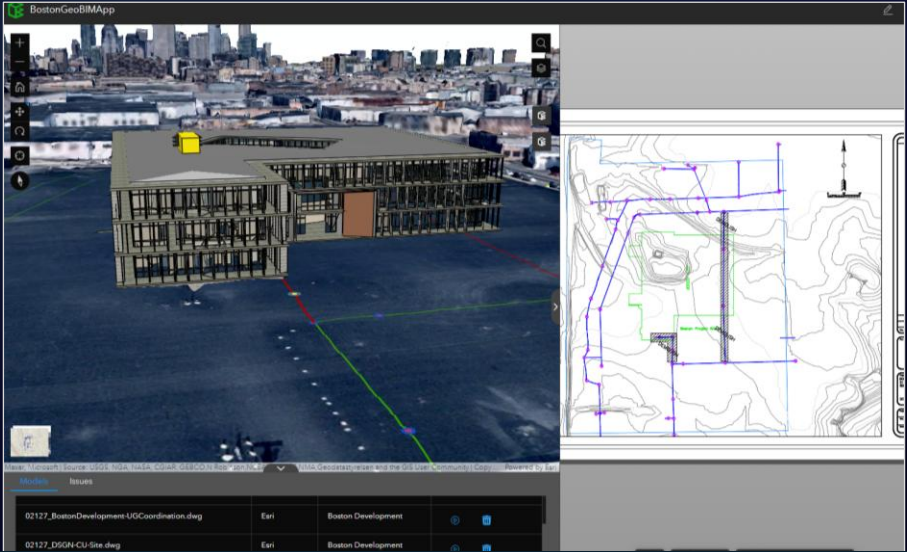
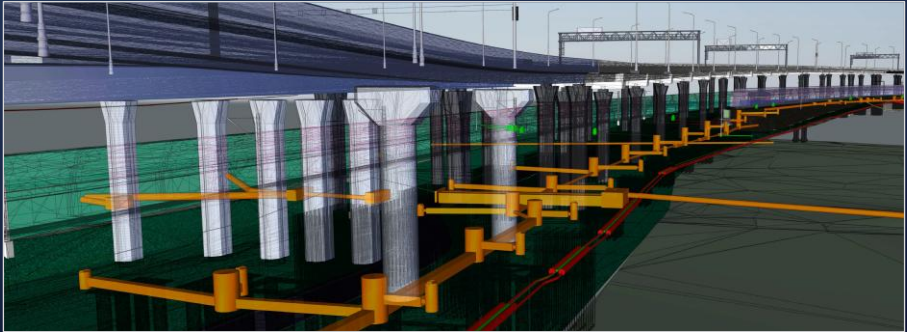
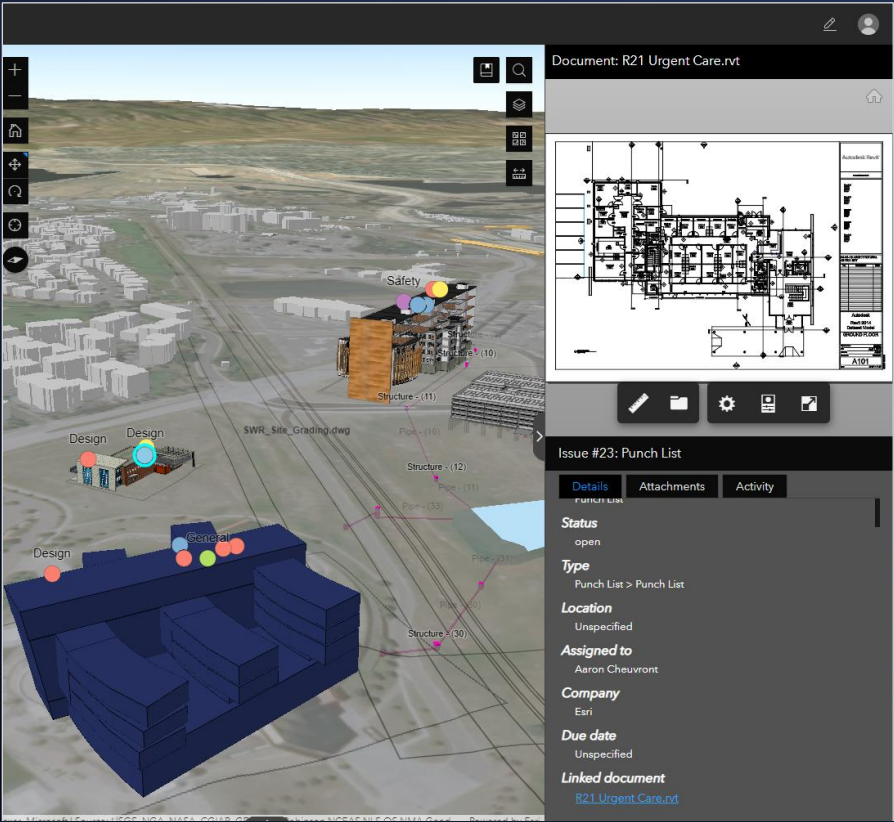
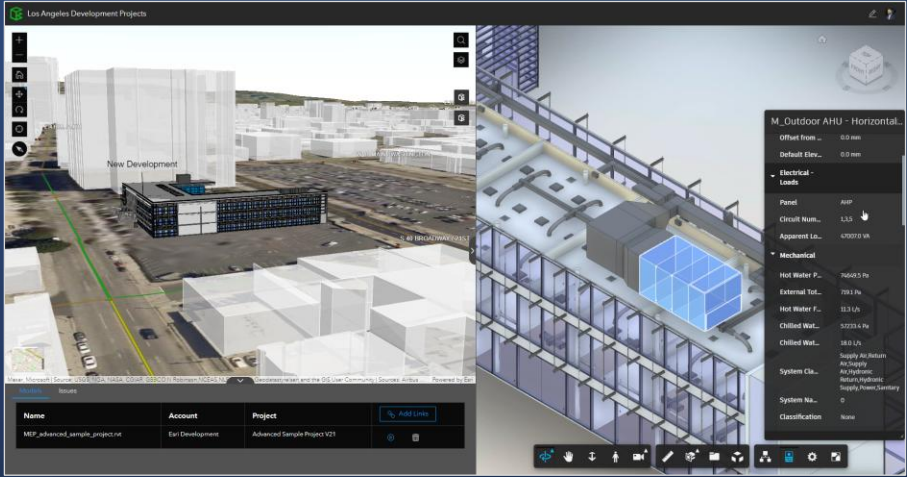
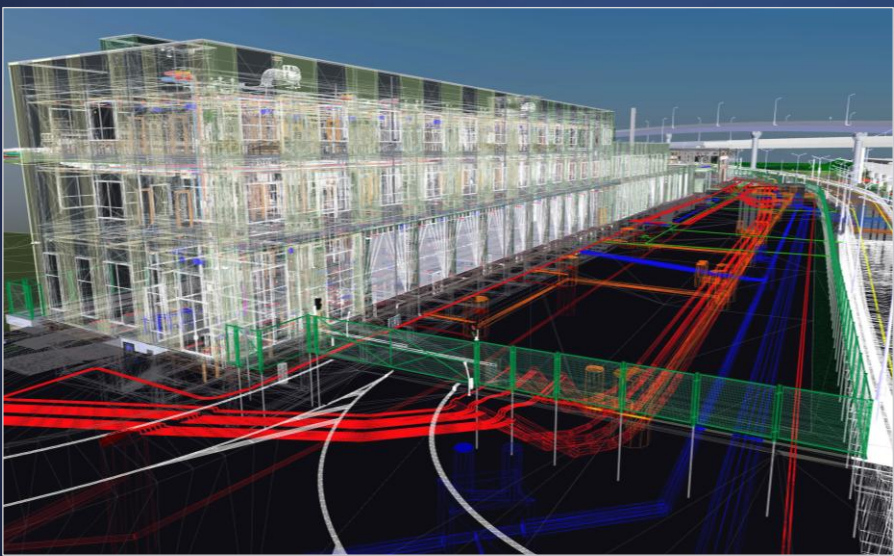






Design & Construction

- Integrates BIM, GIS, schedules, and field data into a unified, accurate project view.
- Improves coordination and reduces risk through spatial clash detection and sequencing.
- Boosts field productivity with real-time updates, inspections, and asset information.
- Optimizes cost, schedule, and sustainability with data-driven insights.
- Enables digital delivery by capturing as-built data in a geospatial twin that seamlessly transitions to operations.



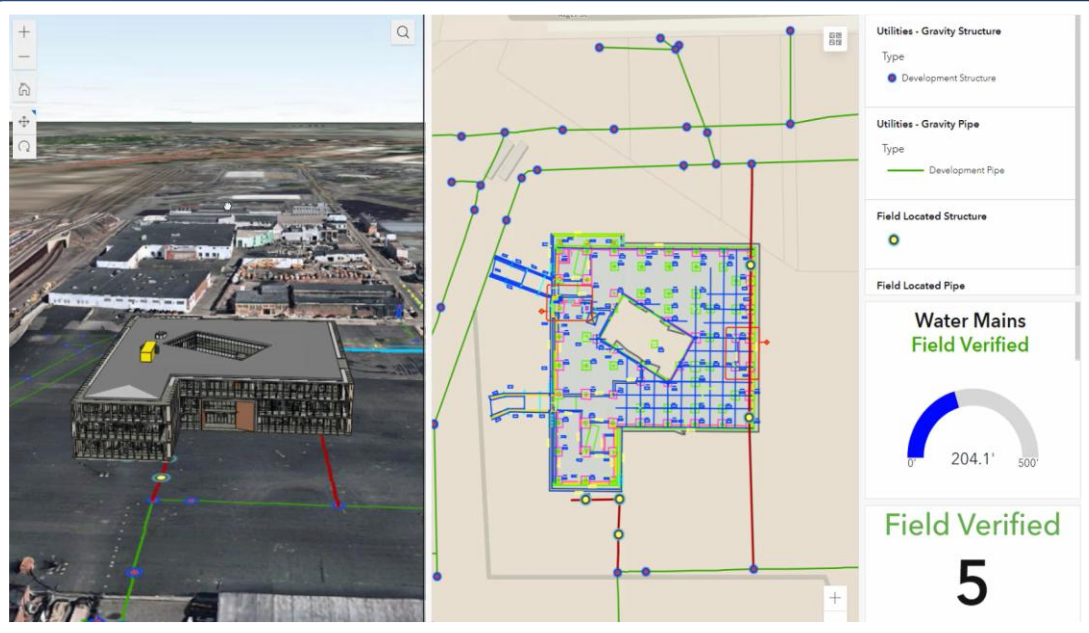
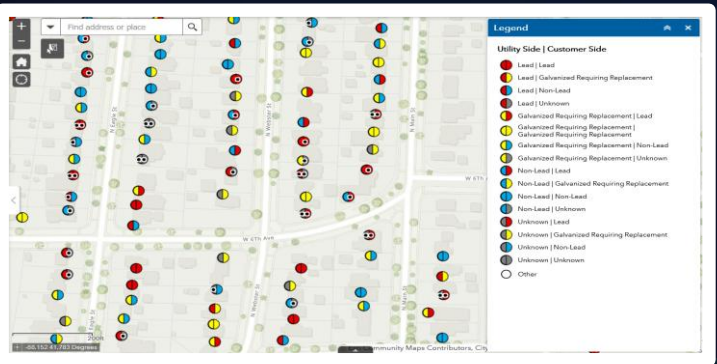
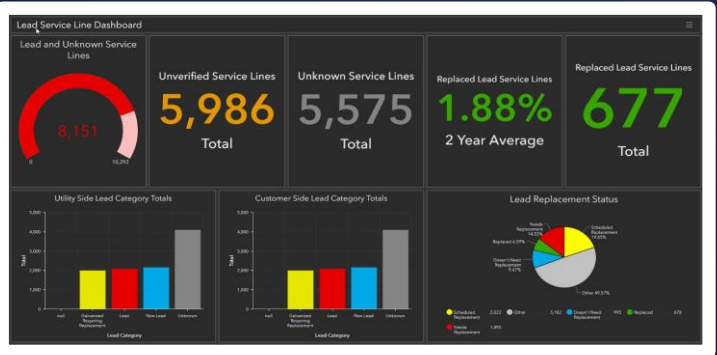
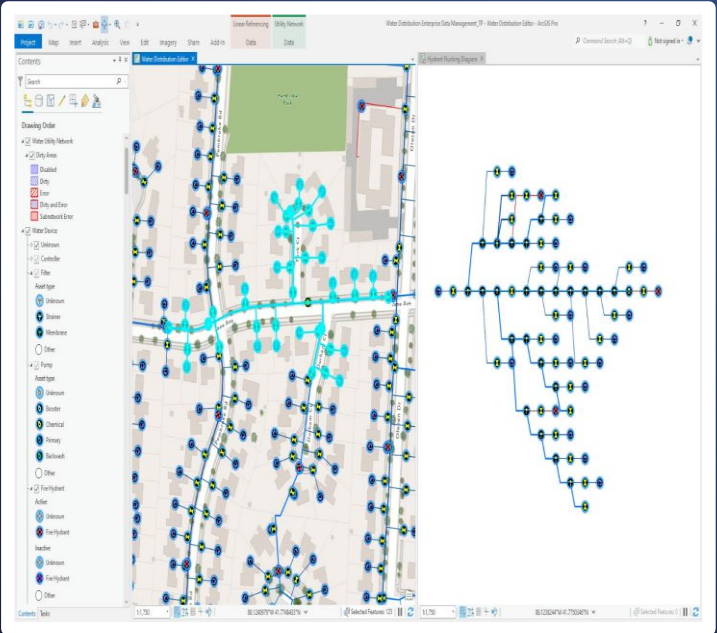


ArcGIS GeoBIM



Network Twin

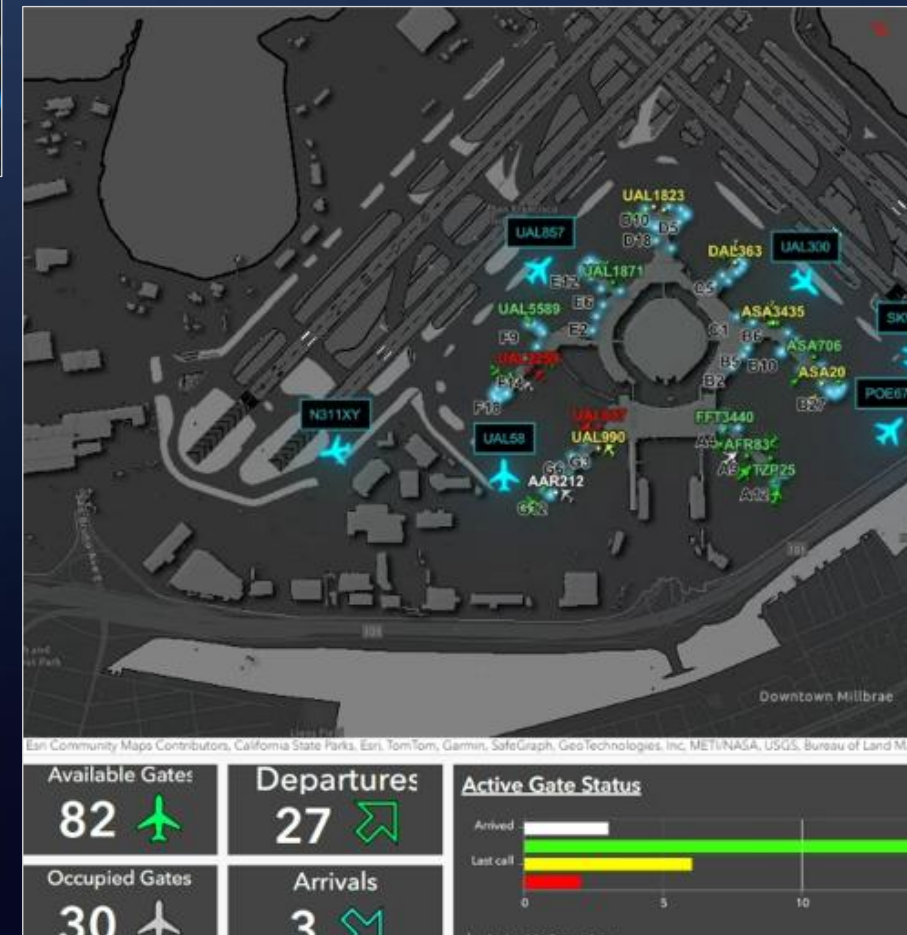
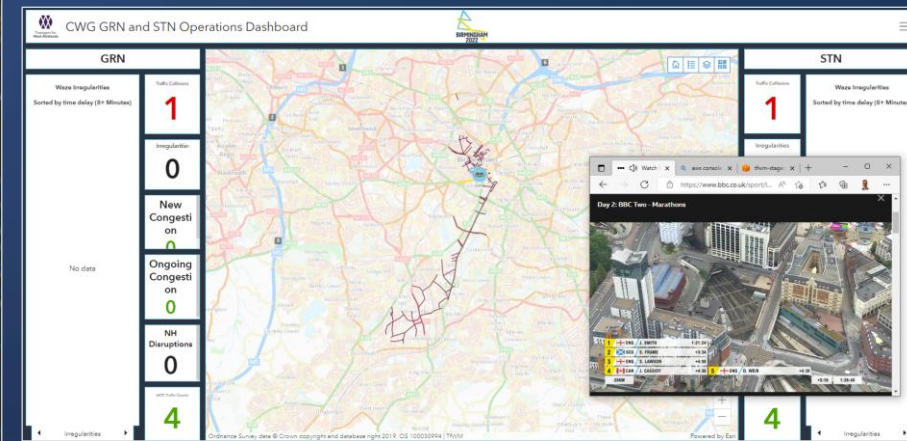
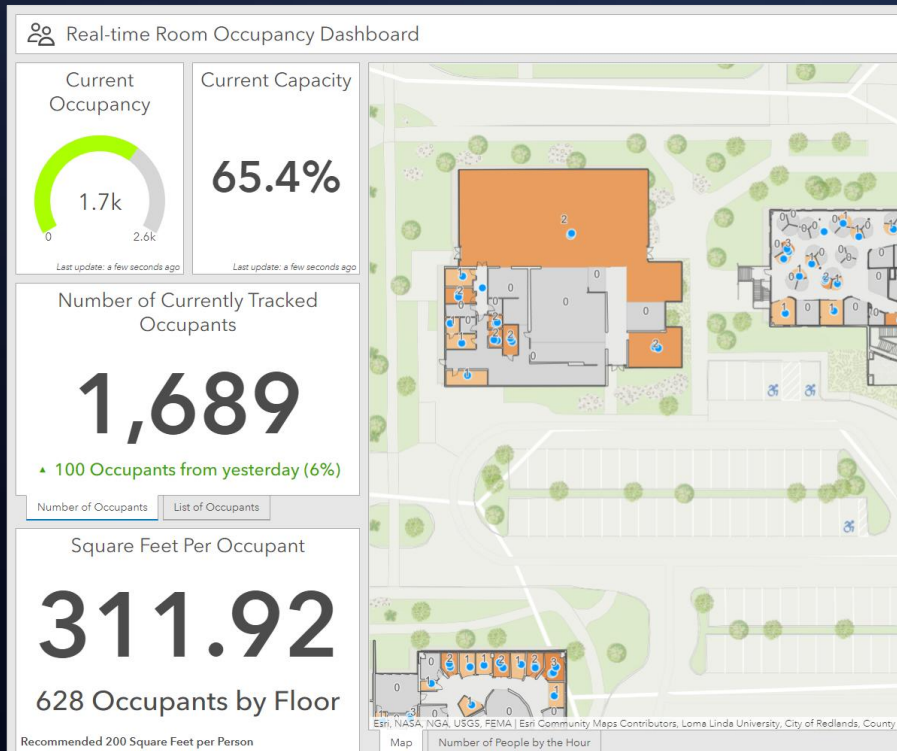
- Model connectivity of linear assets through accurate spatial representation.
- Enables predictive maintenance by integrating IoT, condition data, and risk models.
- Supports operational efficiency by optimizing routing, outage response, load management, and service restoration.
- Reduces downtime with real-time event detection, network tracing, and automated workflows.
- Improves capital planning with spatial analytics on asset performance, replacement cycles, and resiliency.





Operations Twin

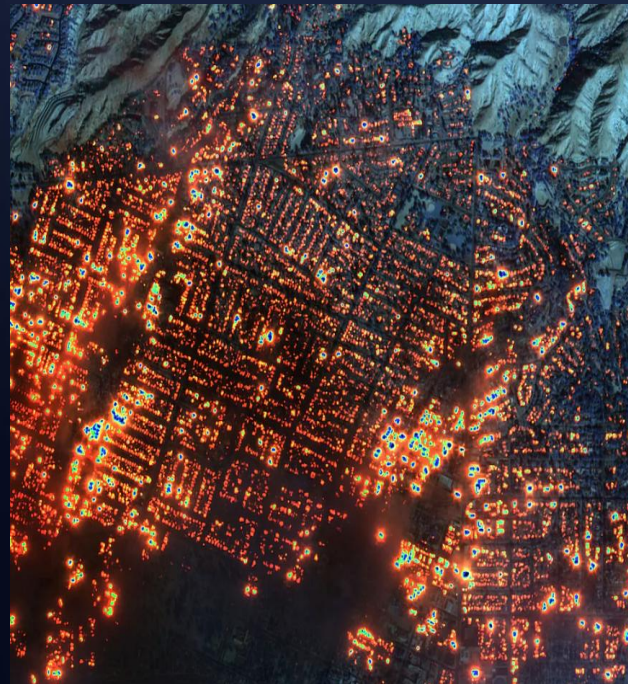
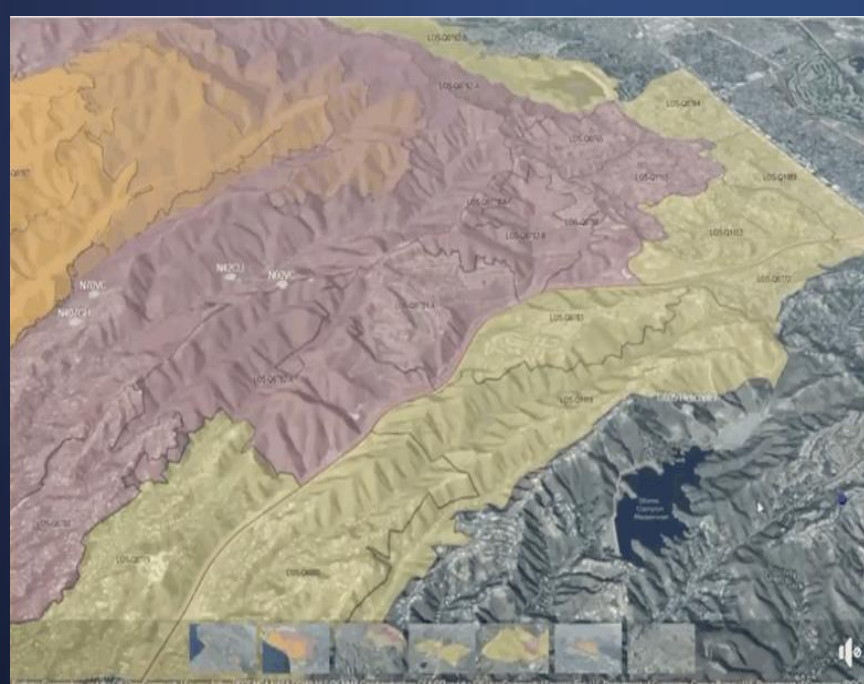
- Delivers real-time visibility into assets and infrastructure for improved situational awareness.
- Supports rapid response and incident management through integrated sensors, cameras, and operational systems.
- Optimizes space utilization, traffic flow, energy usage, and service delivery.
- Breaks down silos by connecting operational systems to a common geospatial frame.
- Drive operational efficiency through predictive insights, simulations, and automated workflows.



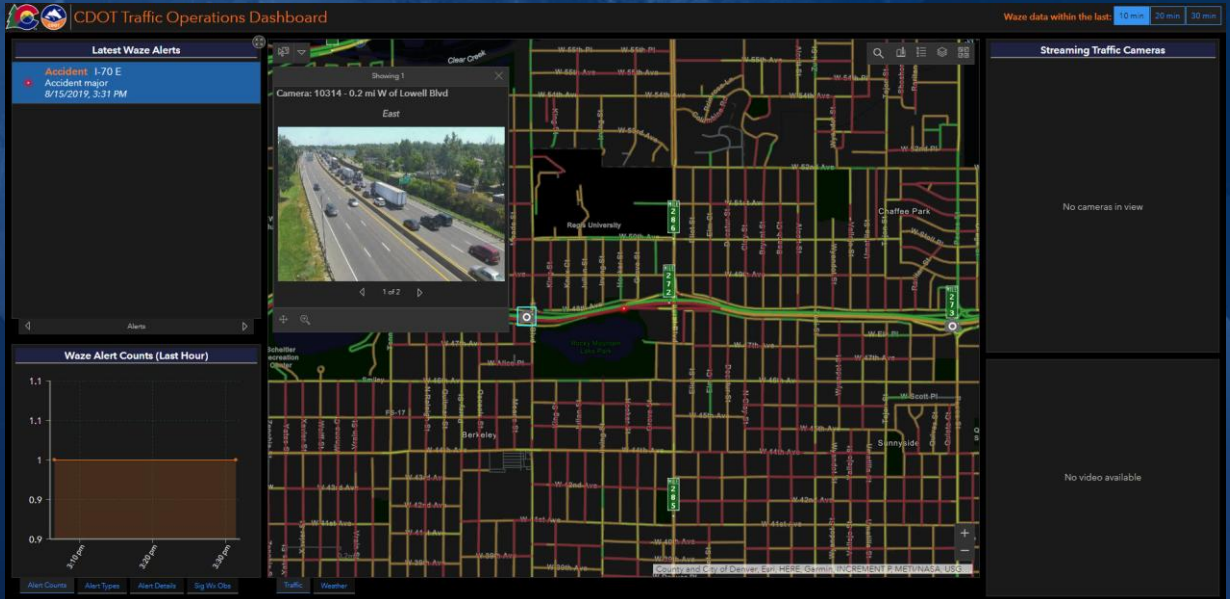
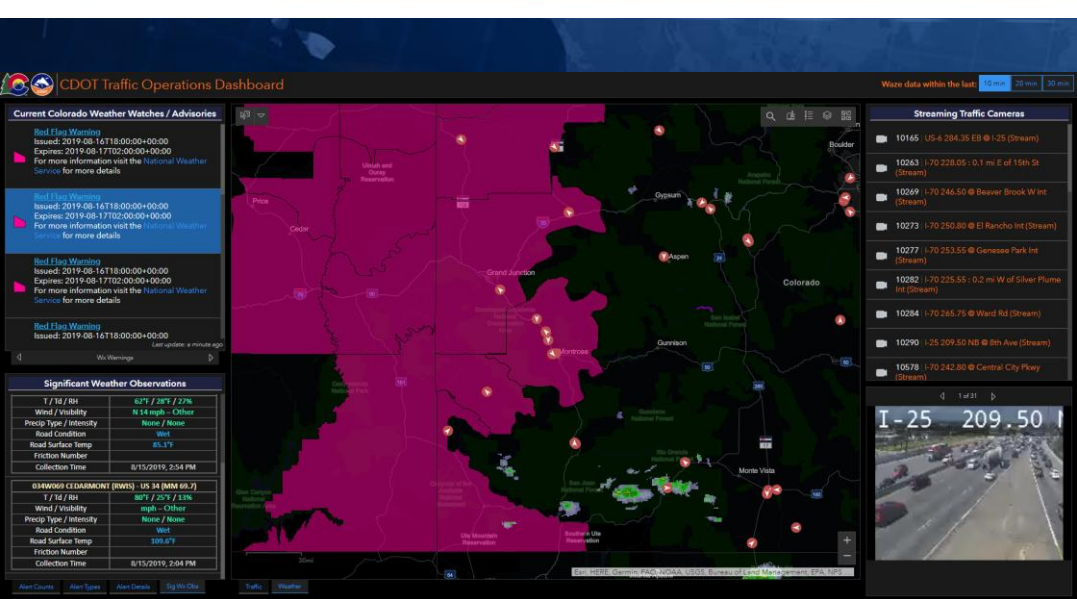
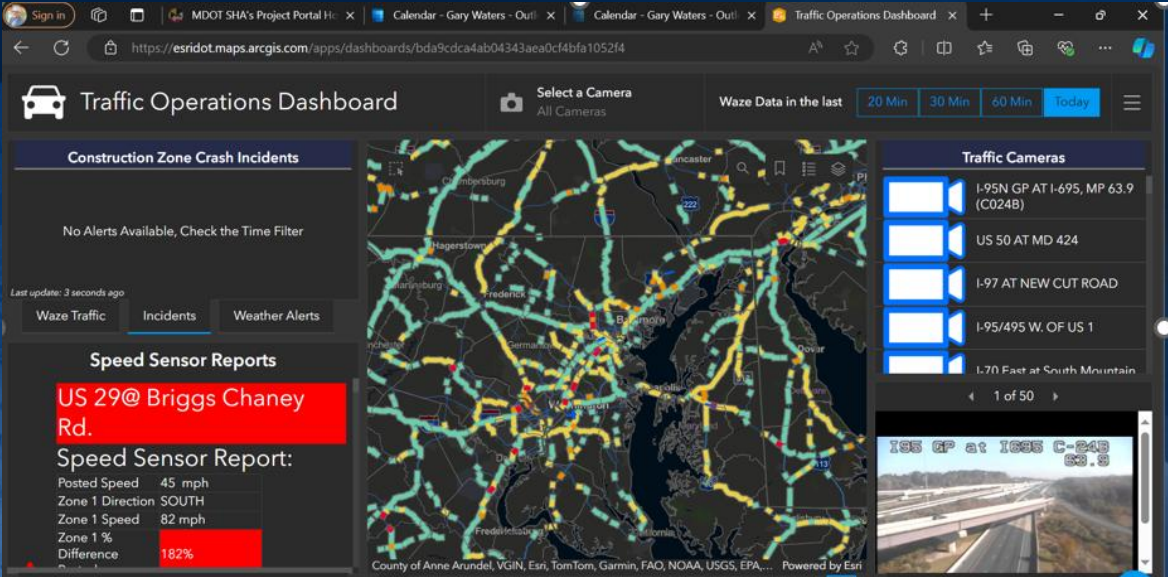
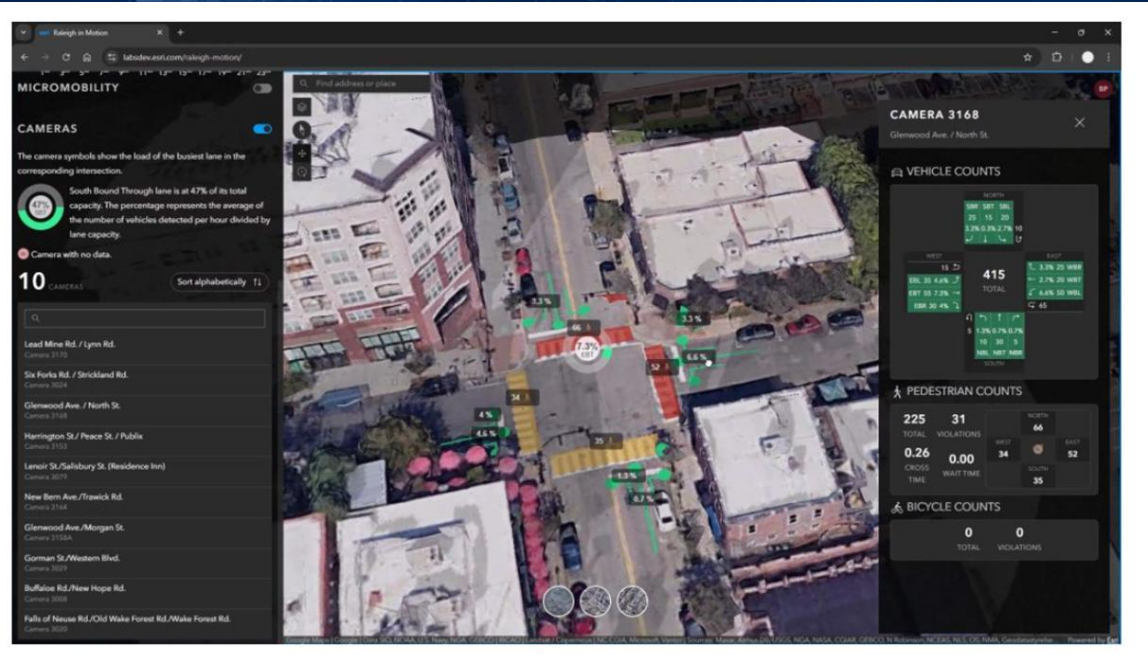


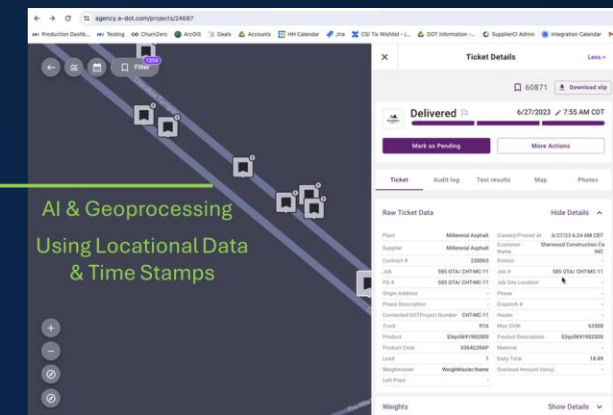
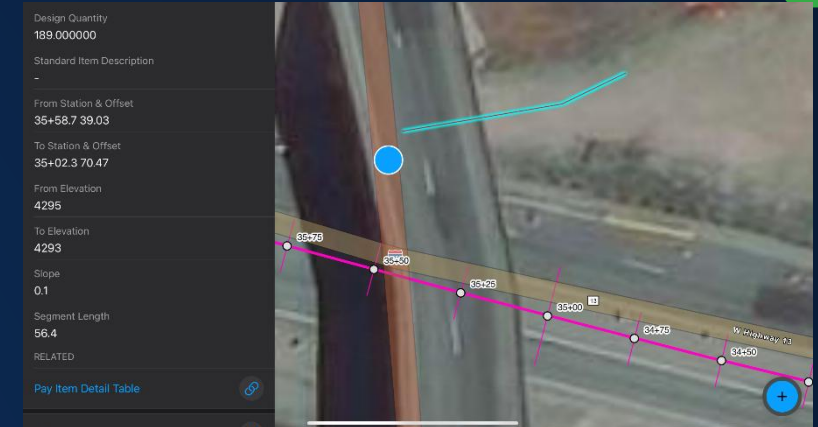
Emergency Management Twin

- Coordinate emergency actions from Planning, Responding, through Recovery.
- Deliver a single, real-time Common Operating Picture.
- Enable Rapid Damage Assessment.
- Apply GeoAI and machine learning to predict risks and optimize resource allocation.
- Integrate environmental, planning, and operations digital twins to simulate scenarios, assess impacts, and improve resilience strategies.
- Facilitate transparent communication and collaboration with customers and communities.



Integrating Sensors and Realtime Data





Project & Workzone Locations for the Public

Sign in

MDOT SHA's Project Portal Home

https://mdot-sha-project-portal-maryland.hub.arcgis.com

Maryland.gov

Home

About

Contact

Project Portal Home

Project Portal Project List

Project Portal Map

Home

Map

Layers

Legend

Full Screen

Print

Share

Map

Layers

Legend

Full Screen

Print

Share

1 of 2

I-68 (National Freeway) bridge over Creek Rd - Bridge replacement

For more information: [Click Here](#)

Phase: Construction

Jurisdiction: Washington

City/Town: Hancock

Work: Bridge replacement

Average Annual Daily Traffic Volume: 18,710

Speed Limit: 70 mph

Street View (if available)

News Release (if available)

[Zoom to](#)

Map

Layers

Legend

Full Screen

Print

Share

Map

Layers

Legend

Full Screen

Print

Share

30mi

POWERED BY

Type here to search

56°F

10:09 AM

10/30/2024

Wildlife Tracking for Collision Prevention in Near-Real Time

Nebraska DOT



NDOT and Nebraska Game and Parks obtained solar-powered GPS collars equipped with remote satellite communication and attached them to seven bighorn sheep. This enabled real-time monitoring of the animals in the study area

Real-time data is not only crucial for preventing wildlife-vehicle collisions, but it can also serve as an early warning system for researchers when radio-collared, wild bighorn sheep interact with other species.

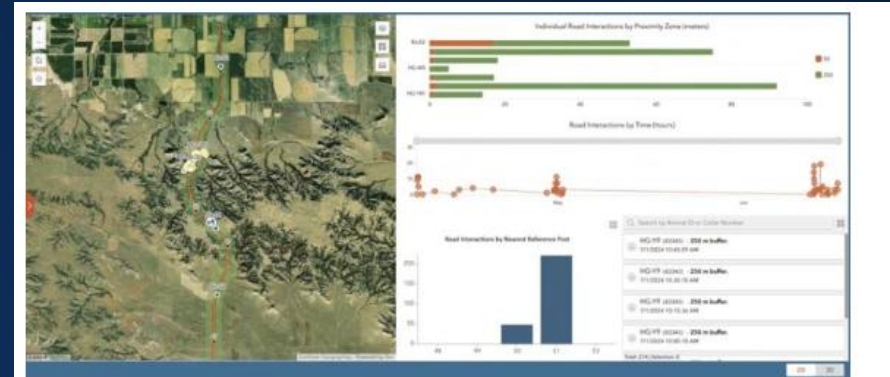
Wildlife Tracking for Collision Prevention in Near-Real Time

Nebraska DOT



A 3D map of a highway segment shows where collared bighorn sheep are and reveals where they often cross roads, indicating prime areas for developing wildlife crossings.

The 3D visualization system notified Nebraska Game and Parks wildlife managers when bighorn sheep were 50–250 meters from a highway.

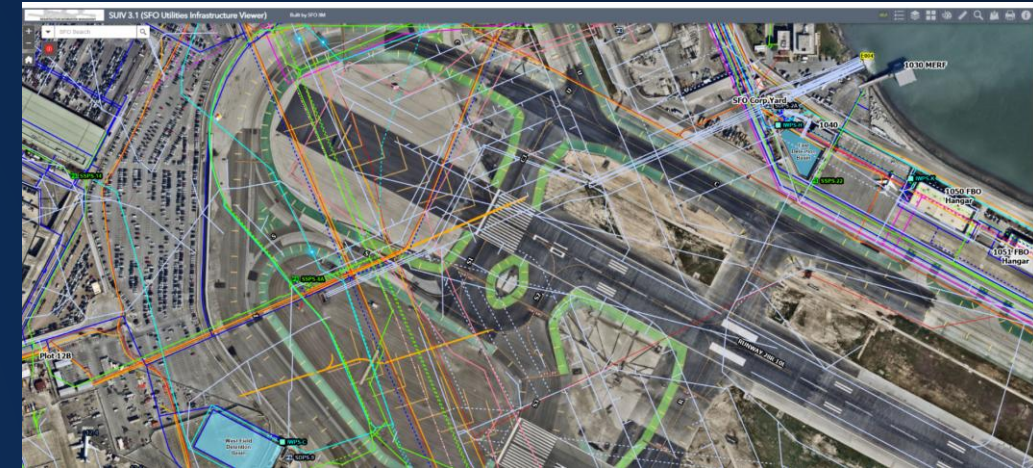
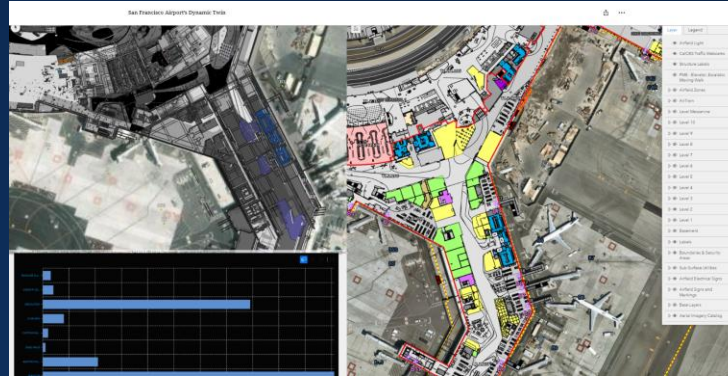


A data visualization dashboard tracks the movements of bighorn sheep in Nebraska, displaying how many are in the vicinity of roads and how often this occurs.

As a result of integrating the GPS collar data with ArcGIS Online and ArcGIS Velocity, wildlife managers were able to proactively monitor bighorn sheep movements in real time to prevent possible wildlife-vehicle collisions and strategically prepare for future wildlife crossings

Dynamic Twin

Development of GIS Viewers



GeoEye 3D GIS Viewer and SFO UIV

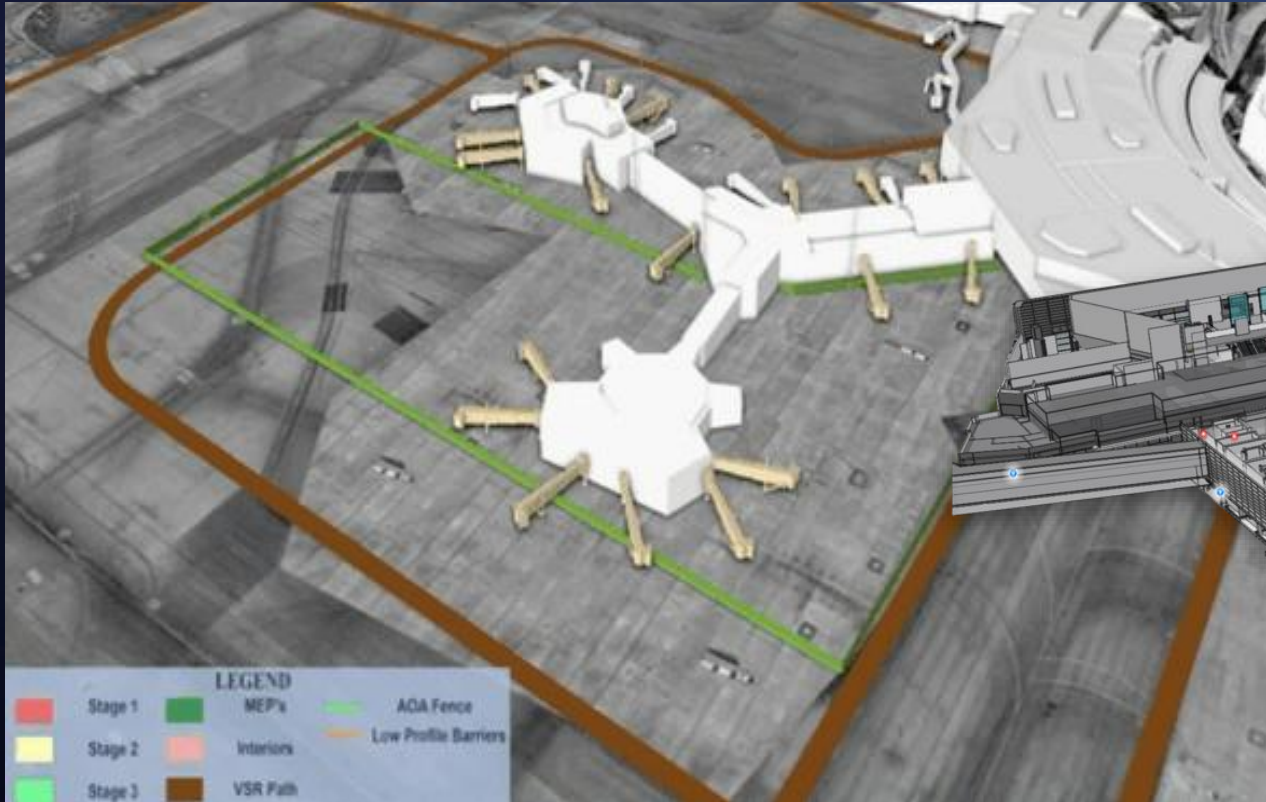
GIS contains >700,000 features
17MM square feet have been mapped
Cad, GIS, BIM and aerial captured in near real time

From Descriptive...

Laying the Foundation: Reality Mapping and Digital Models for Operational Clarity



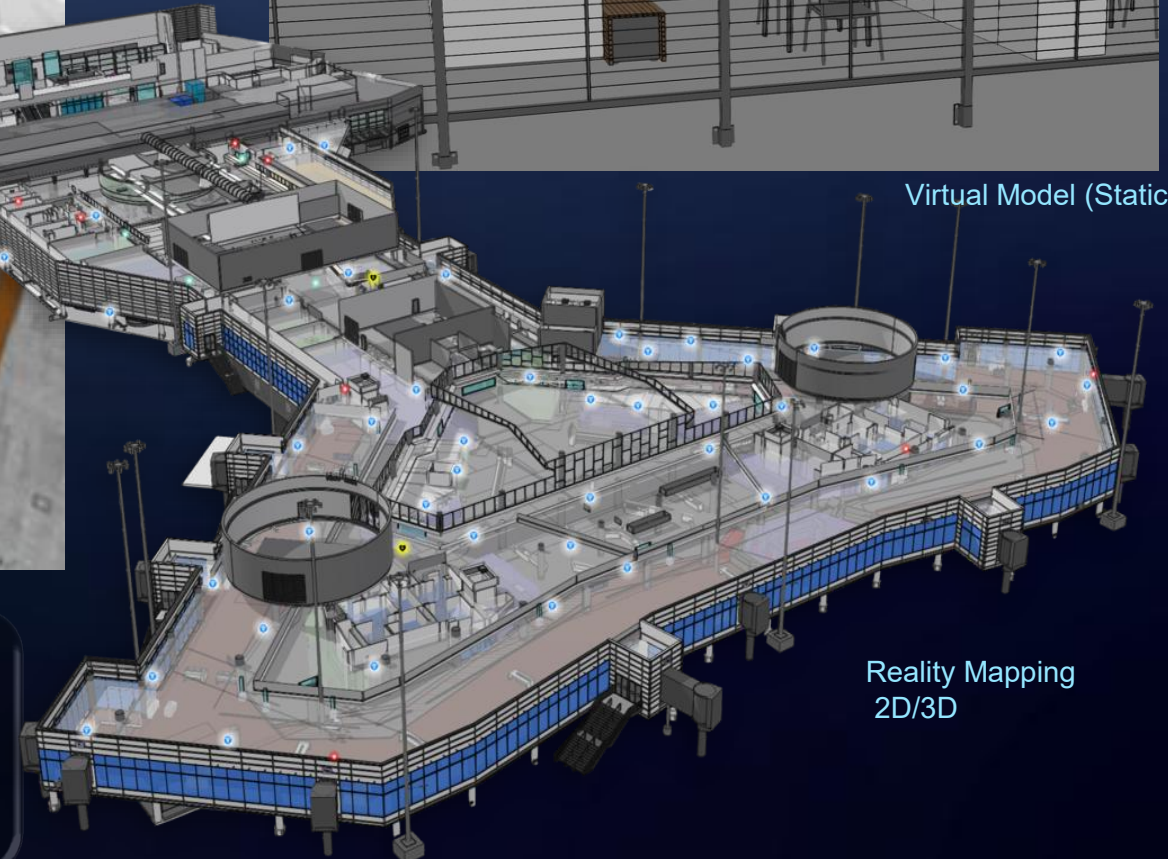
SAN FRANCISCO
INTERNATIONAL AIRPORT



BIM & GIS
Integration



Virtual Model (Static)



Reality Mapping
2D/3D

“A Complete digital transformation of Operations and Asset management with a GIS team of 3”.

Hanson “Guy” Michael (GeoSpatial Systems Principal)

...to Informative and Predictive

Empowering Decision-Making: Dynamic Models, Real-Time Data, and Predictive Insights



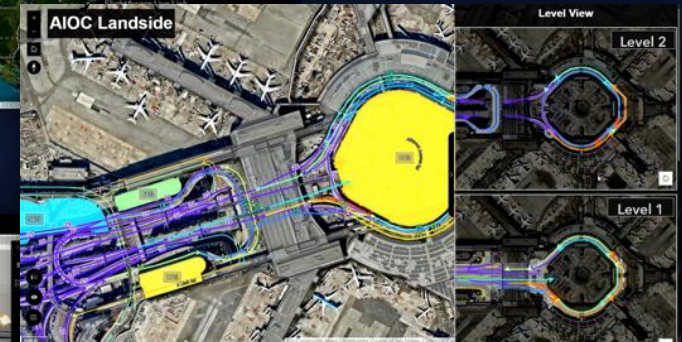
SAN FRANCISCO
INTERNATIONAL AIRPORT



Predictive Modeling



Integrated Operations Center



Virtual Model (Dynamic)

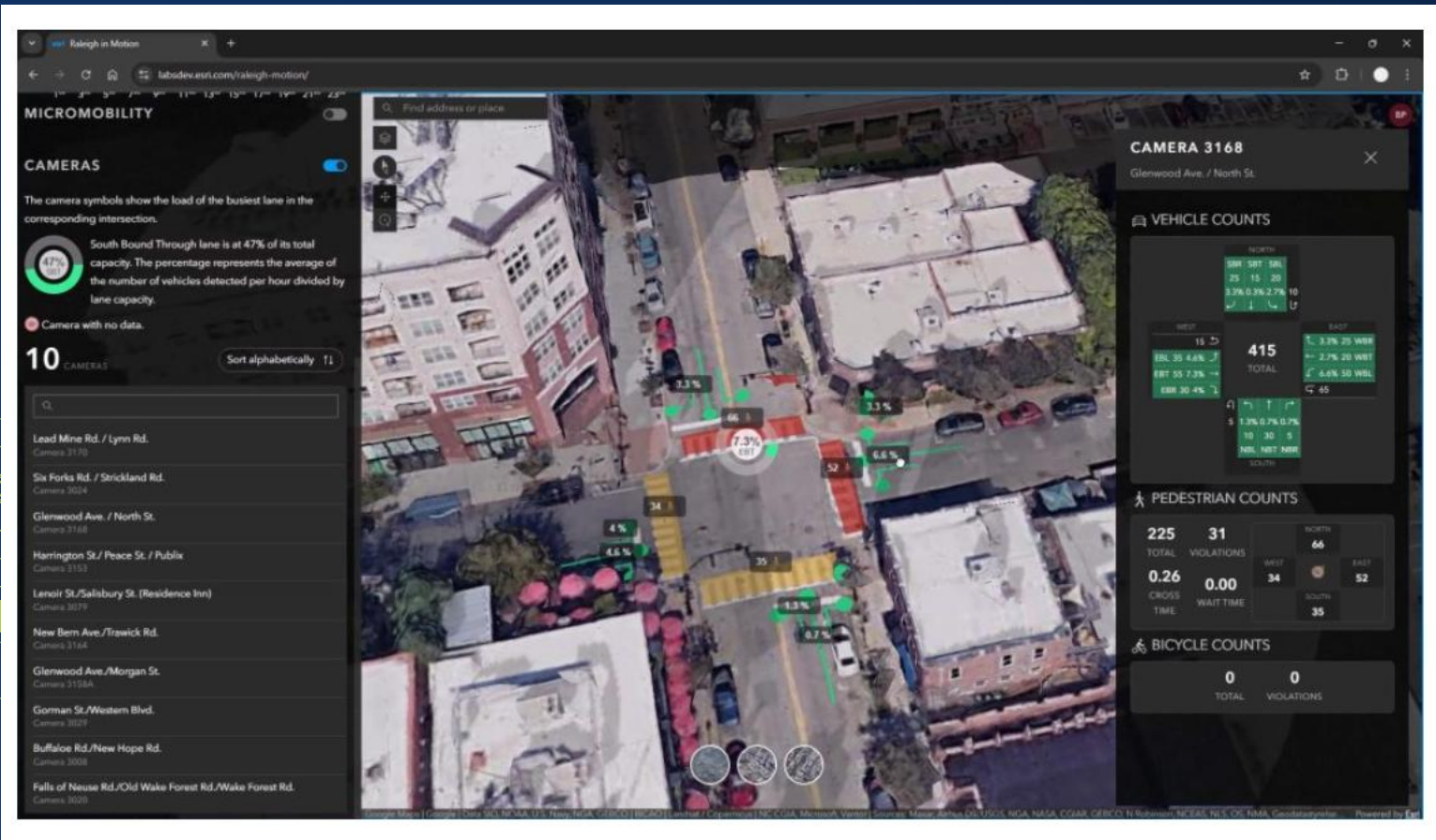


“Every data points has the potential to transform operations, to influence decisions, and to drive innovation”

Josephine Young (Director of Infrastructure Information Management)

Dynamic Operational Twin

City of Raleigh

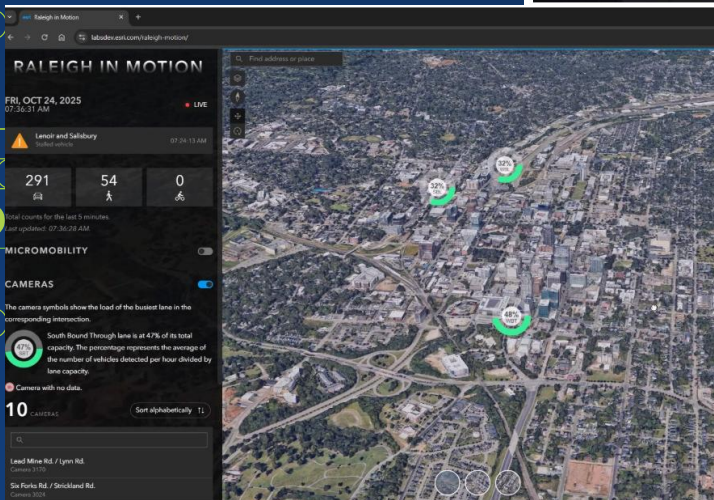
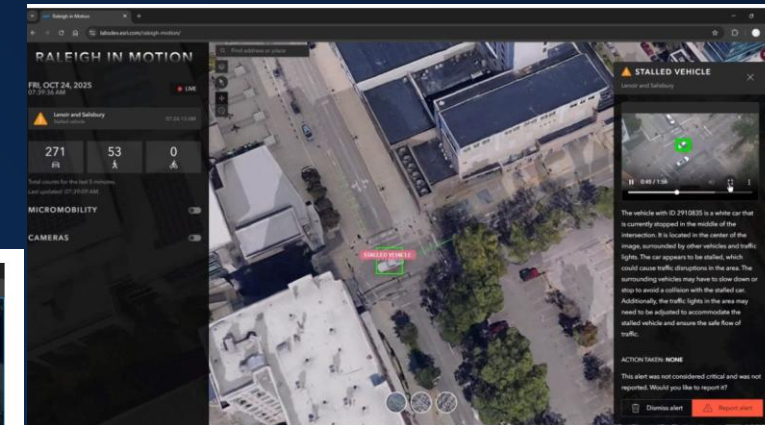
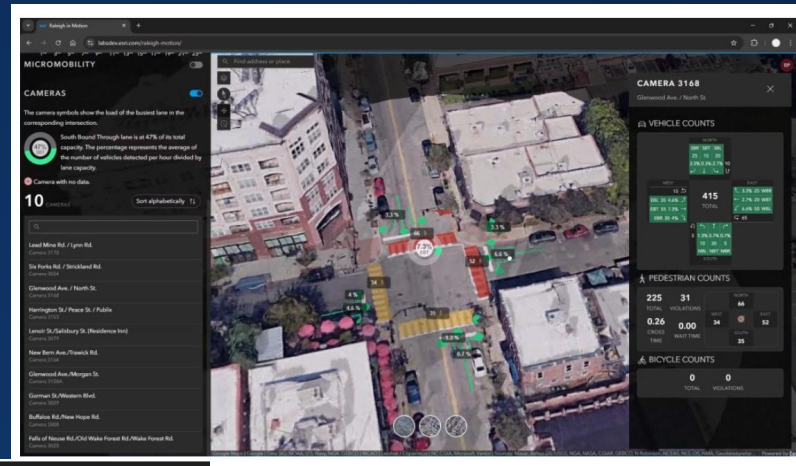


Background

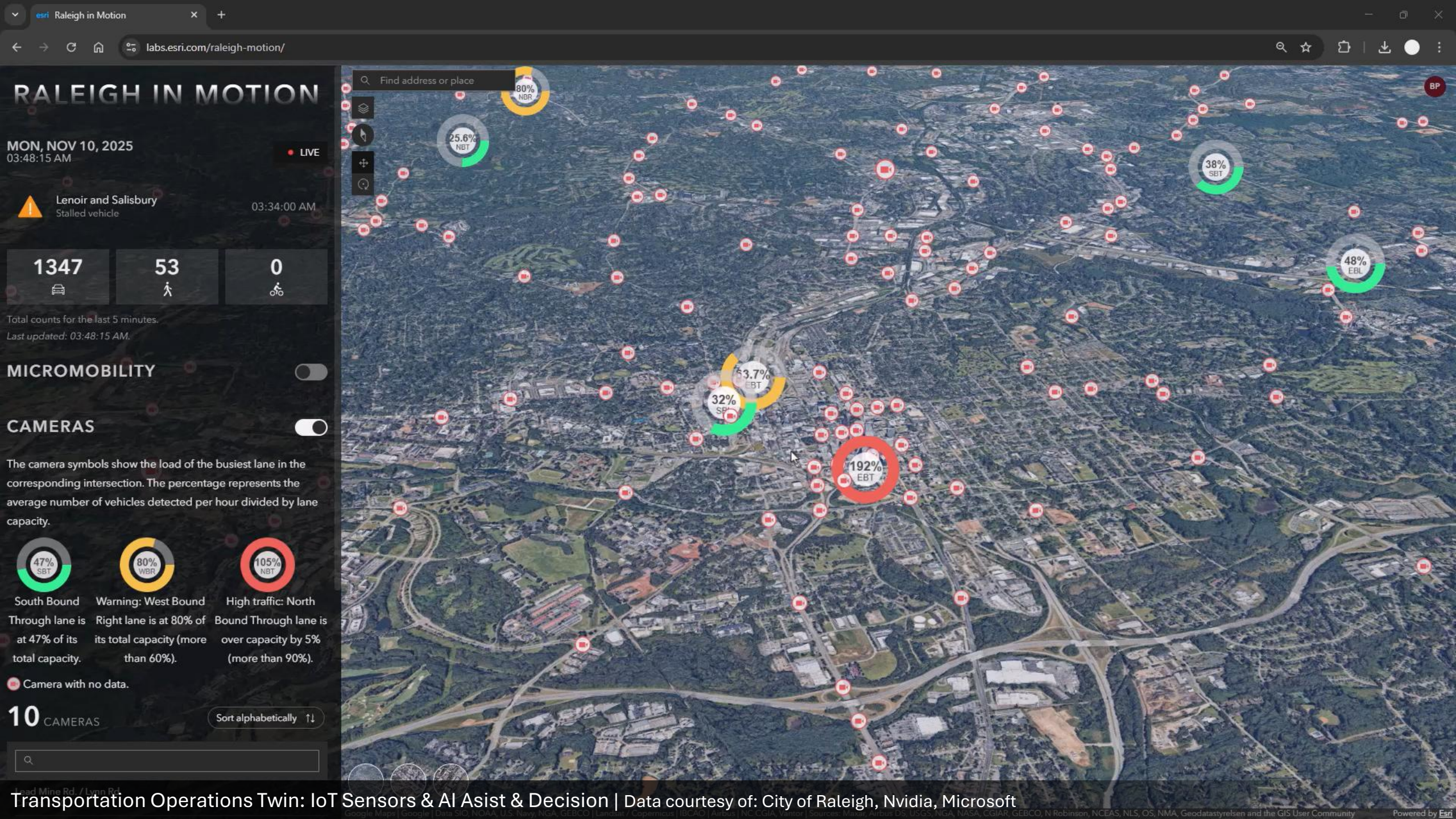
“Raleigh in Motion” pilot leverages Nvidia, Microsoft and Esri ecosystem to monitor traffic flows in real time

Dynamic Twin

City of Raleigh



By fusing real-time streaming data via ArcGIS GeoEvent Server and Velocity, with computer vision capabilities from Nvidia, this digital twin can help Raleigh identify dangerous intersections, reduce congestion and provide safer roadways for vehicles, bikes, and pedestrians.



Digital Twin

A Shared Understanding



- *Location is foundational to every Digital Twin*
- *EGIS connects the social, natural, and built environments*
- *Digital Twins must be outcome-driven*
- *Digital Twin implementation is a journey*

... Creating Living Digital Twins





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